Complete manual

Pico 2 W for the BC250 · power-on board, DualSense dongle and button with LED

How it is organised

Part	Contents	Pages of this file
1 · The board	Building the perfboard from scratch: components, 12 steps, multimeter checks, tests, installation, troubleshooting	2 – 31
2 · The button with LED	Gebildet 12 mm button: 4 wires, 220 Ω resistor on the black wire, checks, MicroPython tests, firmware behaviour	32 – 44

Where the two parts differ, Part 2 applies.
Chapter 7 of Part 1 (firmware) describes the project before the button. The up-to-date pinout and behaviour are in chapter 4 of Part 2: GP8 power-on, GP26 sensor, GP10 button, GP12 LED; a press with the machine off = power on, 4 s = forced power-off, double click = pairing.

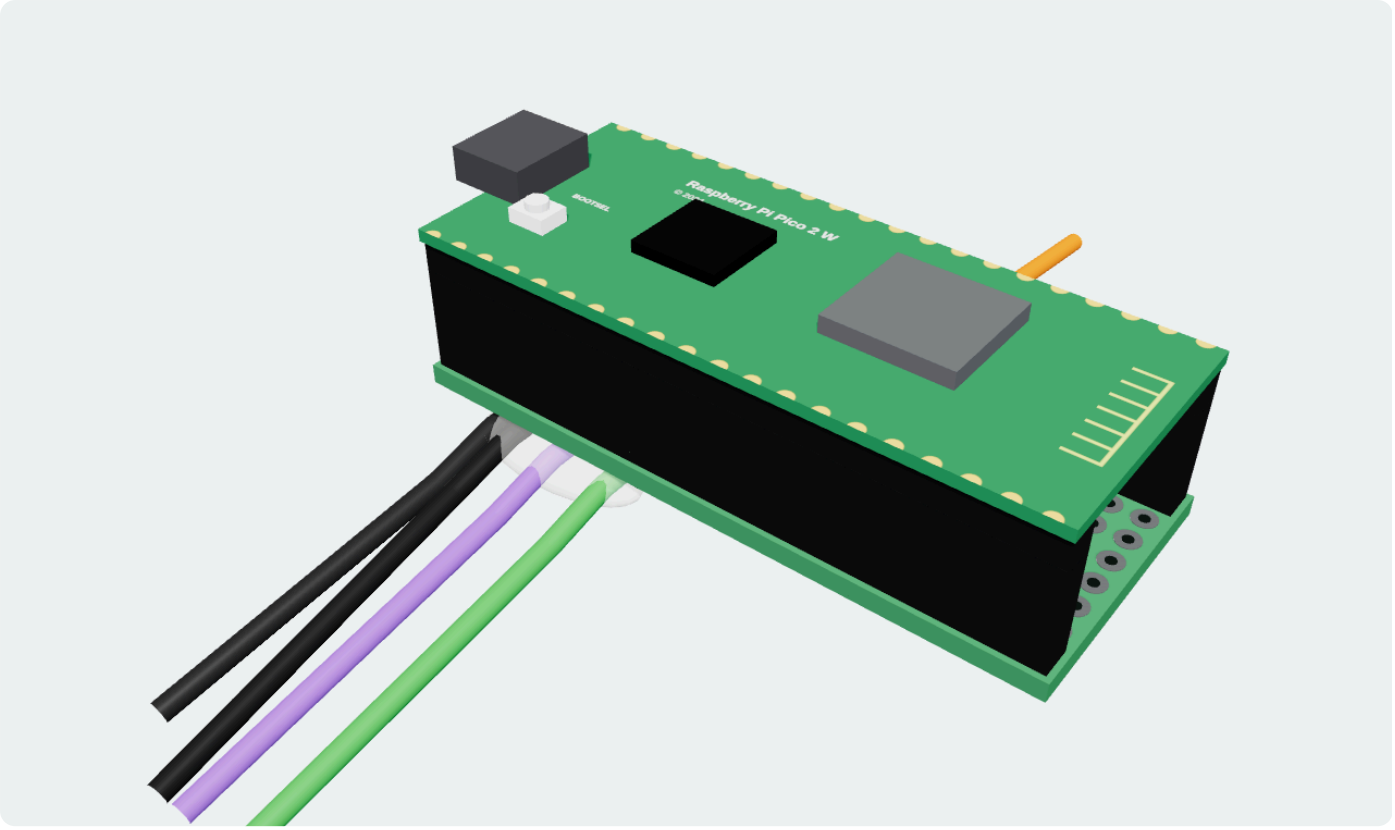
Order of work

1. Part 1, steps 1–12 and Tests 1–3: the board turns the power supply on and off.
2. Part 2, steps 1–7 and tests 1–4: button and LED.
3. Final firmware with the prompt for Claude Code.

English edition of the two guides · 3 October 2026

Assembly manual

Pico 2 W for the BC250 · DualSense dongle and automatic PSU power-on
Compact version on a 20 × 8 hole perfboard, no capacitor



The finished board: the Raspberry Pi Pico 2 WH plugged into its sockets, with every component hidden underneath.

What you get	Turn on the DualSense and the Steam Machine powers up. Shut down from SteamOS and, a few seconds later, the power supply turns off too. Meanwhile the Pico works as the wireless receiver for the controller.
Difficulty	Beginner with a little soldering practice: 6 components (one sits on a wire), about 60 solder joints
Time	2–3 hours of assembly + 30 minutes of testing
Size	about 5.2 × 2.2 cm: the board is as big as the Pico
Version	1.0 · 26 September 2026

Contents and how to use this manual

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In short · Block diagram · Operating cycle · Schematic · Pico pins

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How to read this manual

Each assembly step takes one page: at the top there is a **3D render** of the result, next to it the **numbered instructions**, below the **exact hole diagram**, with what you do in that step highlighted in yellow. Follow the steps in order: they are arranged so you never have to flip the board more than necessary.

> **Warning**
> These are the points where a mistake can damage something or make the circuit misbehave.

> **Check**
> A check to do right away, usually with the multimeter, before moving to the next step.

> **Tip**
> A practical trick that makes the job easier or cleaner.

The three golden rules

1. **Always work with the power supply unplugged from the wall.** The 5V standby rail is live even when the fans are not spinning.
2. **Check with the multimeter before applying power.** Every step has its own check: doing them all takes a few minutes and saves you from burning something.
3. **The pins marked with a red ✗ get only the header solder joint**: VBUS, EN, 3V3, VREF and RUN. No wire, no bridge, no solder reaching the neighbouring holes.

1 What you build and how it works

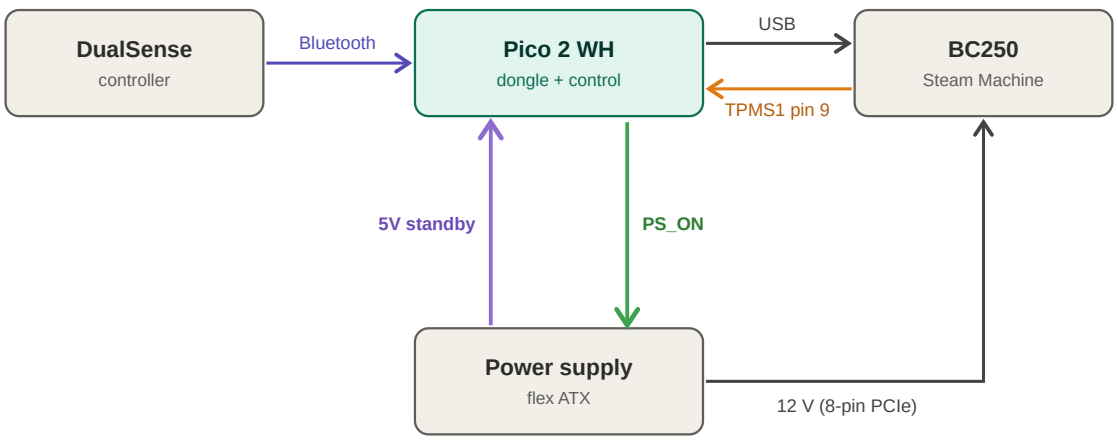
1.1 In short

The **BC250** has no power button: it starts by itself as soon as it gets 12 V from the power supply. Usually people jumper the green PS_ON wire of the PSU and switch everything from the mains switch. This board replaces that jumper with a **Raspberry Pi Pico 2 W** that:

- works as the **wireless receiver for the DualSense** (DS5Dongle firmware): the BC250 sees it as a wired controller;
- **turns the power supply on** when you turn on the controller, by pulling the green PS_ON wire to ground through an optocoupler;
- **knows when the BC250 has shut down** by reading a pin on the board's TPMS1 header, and releases PS_ON after 3 seconds.

The Pico is always on because it is powered from the **5V standby** rail (purple wire), which the PSU keeps live as long as it is plugged in.

1.2 Block diagram

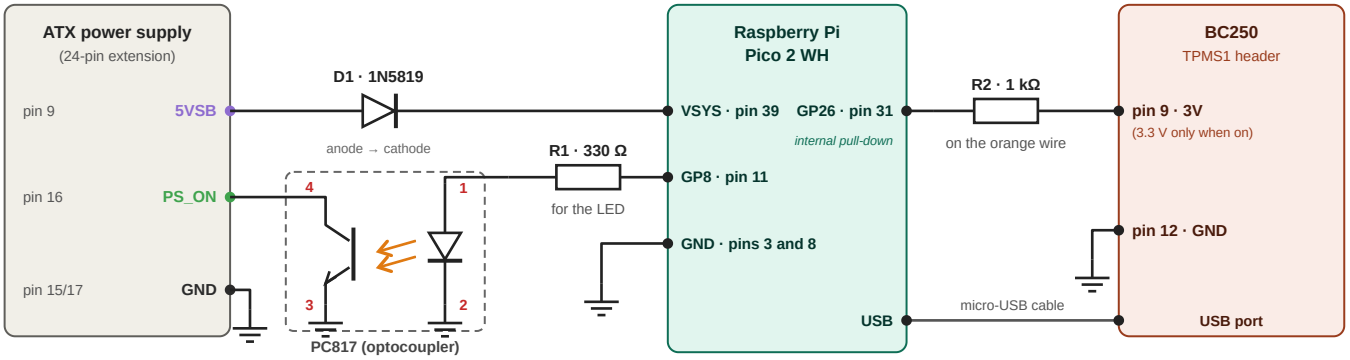


Who talks to whom. The Pico is the only part that is always on.

1.3 The operating cycle

1. Machine off: the PSU is in standby, the Pico is powered from 5V standby and waits.
2. You turn on the DualSense: the controller connects to the Pico, which drives GP8 high. The LED inside the PC817 lights up, its transistor connects the green wire to ground and the PSU starts.
3. The BC250 gets 12 V and boots by itself. Pin 9 of TPMS1 rises to 3.3 V: the Pico reads it on GP26.
4. You play: the Pico acts as the dongle and checks the sensor every 100 ms.
5. You shut down from SteamOS: the BC250 turns off and pin 9 drops to 0 V. After 3 seconds the Pico drives GP8 low and the PSU goes back to standby.

1.4 Schematic



All the ground symbols are the same point: the PSU black wire, the Pico GND pins and pin 12 of TPMS1.

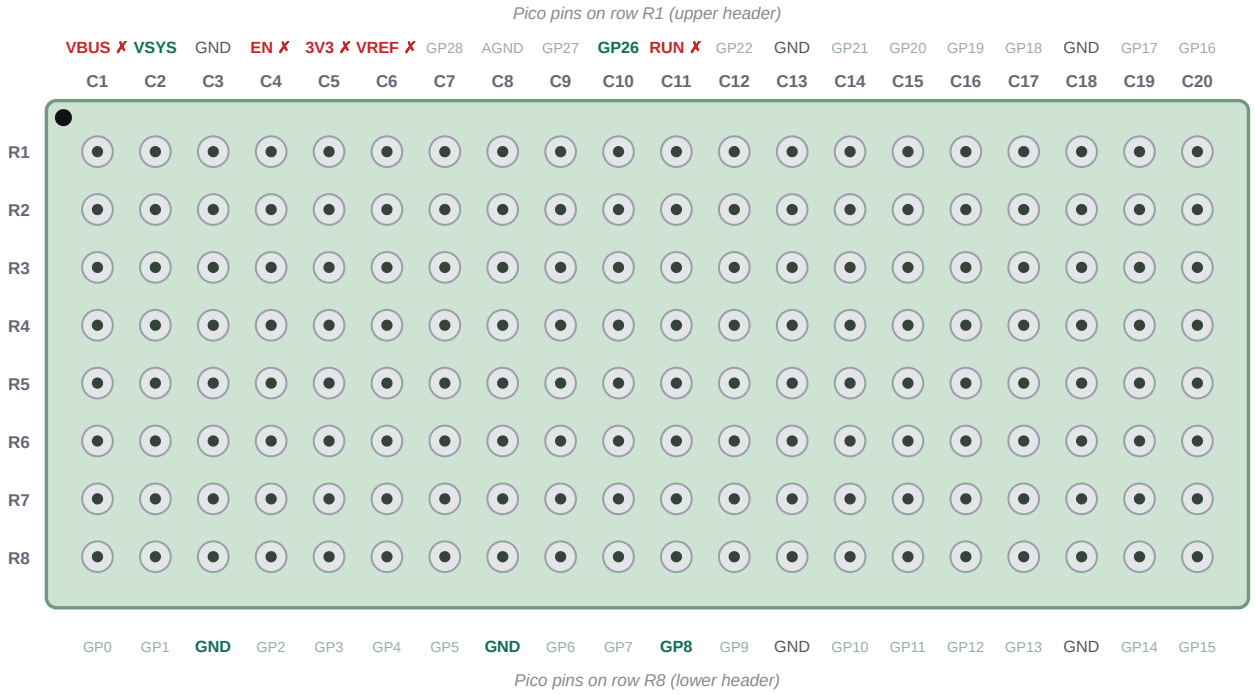
The full schematic. D1, U1 (PC817), R1 and the headers sit on the perfboard; R2 is soldered in the middle of the orange wire.

What each component is for

Ref	Component	What it does	Why this one
D1	1N5819 Schottky diode	Feeds 5V standby to the Pico's VSYS pin.	When USB is also connected, the diode stops the two supplies from fighting each other. A Schottky drops only about 0.3 V: the Pico gets about 4.7 V, and it accepts 1.8 to 5.5 V.
U1	PC817 optocoupler	A light-controlled switch: the Pico lights the internal LED (pins 1–2), the phototransistor (pins 4–3) connects PS_ON to ground.	The green PS_ON wire sits at about 5 V: the PC817 keeps it away from the Pico pin, which only tolerates 3.3 V. If the Pico is off, the LED is off and the PSU stays in standby; if the Pico reboots while the BC250 is on, the PSU turns off.
R1	330 Ω resistor	Limits the current of the PC817 LED.	(3.3 V − 1.2 V) ÷ 330 Ω ≈ 6 mA: enough to drive the PC817 well, little for the Pico pin.
R2	1 kΩ resistor (on the orange wire)	Protects GP26 and the BC250.	If a firmware bug turned GP26 into an output, the current would be 3.3 mA at most.
—	GP26 internal pull-down	Holds GP26 at zero when the BC250 is off.	Enabled by the firmware: no extra resistor needed.
J1, J2	1 × 20 female headers	Hold the Pico and make it removable.	8.5 mm tall: they leave room for the components under the Pico.

1.5 Pico pins used and forbidden pins

Seen from the component side, the Pico sits on the perfboard with the USB socket towards the left edge (column 1). Its 40 pins land on rows R1 (pins 40…21) and R8 (pins 1…20).



Top view of the empty perfboard with the Pico pin names: green = used, red ✗ = forbidden.

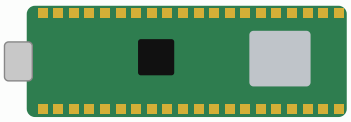
Hole	Pin	Name	Use
C2-R1	39	VSYS	Pico power, from diode D1
C10-R1	31	GP26	BC250 sensor input (through the 1 kΩ R2)
C11-R8	11	GP8	output that drives the PC817 (through the 330 Ω R1)
C3-R8	3	GND	ground: black wires (ATX and TPMS1)
C8-R8	8	GND	ground: PC817 (pins 2 and 3)

Forbidden pins

C1-R1 **VBUS** · C4-R1 **EN** (3V3_EN: pulling it to ground turns the Pico off) · C5-R1 **3V3** (3.3 V output) · C6-R1 **VREF** · C11-R1 **RUN** (pulling it to ground holds the Pico in reset). These holes get only the header pin solder joint: no wire, no bridge, no solder towards the neighbouring holes. Note: C8-R1 is **AGND**, the analog ground: it is not used in this project.

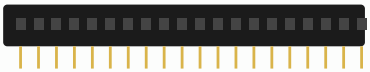
2 Parts, tools, safety

2.1 Components



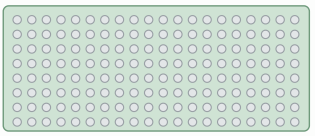
Raspberry Pi Pico 2 WH
with pre-soldered pins

Pico 2 WH · **1**



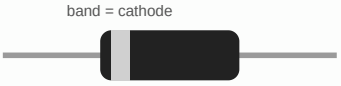
1 × 20 female header
2.54 mm pitch · 8.5 mm tall · you need 2

1×20 female header · **2**



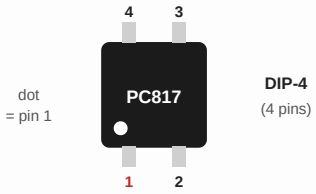
Perfboard cut to 20 × 8 holes
about 5.2 × 2.2 cm including the edge

Perfboard · **1 (to be cut to 20 × 8)**



1N5819
Schottky 1 A, 40 V · DO-41

1N5819 diode · **1**



PC817 DIP-4 · **1**



330 Ω
orange · orange · brown · gold

330 Ω resistor · **1**



1 kΩ
brown · black · red · gold

1 kΩ resistor · **1**



Red solid-core wire
rigid, single conductor · for the connections underneath

Red solid-core wire · **about 15 cm**



2 female 2.0 mm jumper leads
orange (pin 9) and black (pin 12) of TPMS1

2.0 mm jumper leads · **2**



24-pin ATX extension (or breakout)
you need: purple, black, green

ATX extension / breakout · **1**



Heat-shrink tubing
2–5 mm diameters

Heat-shrink tubing · **about 40 cm**



Micro-USB data cable
ordinary, unmodified

Micro-USB data cable · **1**

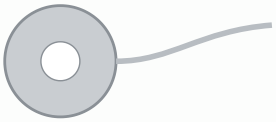
Where to find them

Resistors, diode, PC817, headers and heat-shrink are sold by electronics component shops or in small packs online. The Pico 2 WH, the ATX breakout and the 2.0 mm jumper leads are best bought online. If you can only find a 1N4007 diode it still works, but the values change: the Pico gets about 4.3 V instead of 4.7 and in diode test you will read 0.5–0.7 V instead of 0.15–0.35 V.

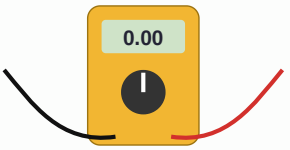
2.2 Tools



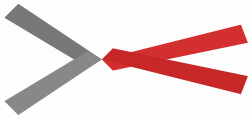
Adjustable soldering iron
320–350 °C · conical or fine chisel tip



Flux-cored solder
0.6–0.8 mm diameter



Multimeter
with continuity (beep) and diode test



Flush cutters and wire strippers
to trim leads and strip wire



Utility knife + metal ruler
to cut the perfboard · file or sandpaper



Hot glue gun
to secure the wires (zip ties as an alternative)



Heat gun or lighter
to shrink the tubing



Helping hands (recommended)

Also useful: desoldering braid, tweezers, magnifier, permanent marker, Kapton tape, safety glasses.

2.3 Safety

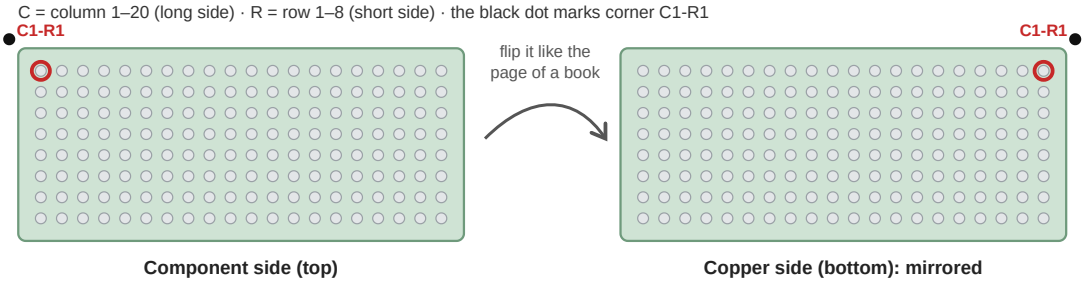
- **Soldering iron:** the tip reaches 350 °C. Always rest it on its stand and never leave it on unattended.
- **Fumes:** flux smoke is an irritant. Work in a ventilated room and do not breathe it directly.
- **Leaded solder:** if you use it, wash your hands afterwards and do not eat while soldering.
- **Glasses:** wear safety glasses when trimming component leads: the offcuts can fly off. Also keep a finger over the piece.
- **Cutting the perfboard:** fibreglass (FR4) makes a fine, irritating dust. Score it instead of sawing, wear a mask and clean up with a damp cloth, not by blowing.
- **Power supply:** never open it. All the work is done on the output wires, at low voltage, and always unplugged.

3 The basics

3.1 Coordinates and the two sides of the board

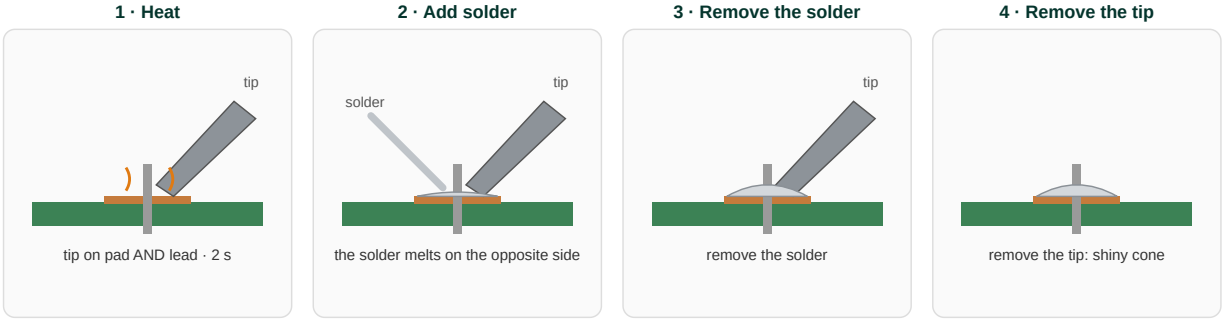
Every hole has a name: **C** is the column (1 to 20, along the long side), **R** is the row (1 to 8, along the short side). For example **C7-R3** is the hole in column 7, row 3.

Components go in from the **component side** (top) and are soldered on the **copper side** (bottom). When you flip the board like the page of a book, row R1 stays at the top but left and right swap: column 1 moves to the right. That is why the copper-side diagrams are **mirrored**, exactly as you will see the board in your hand.



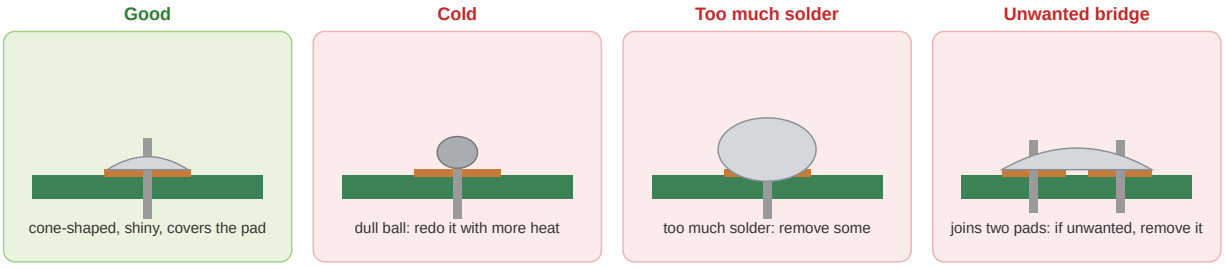
The black dot marks corner C1-R1: draw it with a marker on both sides.

3.2 How to make a good solder joint



The four beats of a solder joint: 2–3 seconds in all.

1. Set the iron to **320–350 °C** and wait until it is up to temperature.
2. Clean the tip on the damp sponge or brass wool, then put a thin coat of solder on it: a shiny tip transfers heat better.
3. Rest the tip so that it touches **the pad and the lead at the same time**. Count to 2.
4. Bring the solder wire in from the side opposite the tip, not onto the tip: it must be the heat of the pad that melts it.
5. When the solder has covered the pad, remove the solder wire first and then the tip. Do not move the part for 2–3 seconds while it cools.



How to recognise a good joint and the three most common mistakes.

3.3 Solder bridges and wires on the copper side

On a perfboard every hole is isolated: you make the connections yourself, on the copper side, in two ways.

The solder bridge (between two neighbouring holes)

1. Solder the two holes normally, each with its component or wire.
2. Heat the first joint and add a little solder, pushing it towards the neighbouring hole.
3. Move to the second joint and do the same until the two blobs merge.
4. If the bridge will not form, lay a piece of leftover component lead between the two holes and solder it at both ends: it is easier and stronger.

The red wire (for longer connections)

1. Measure the run on the board: 2.54 mm per hole of distance.
2. Cut the red wire as long as the run plus 2 mm and strip 2 mm at each end: the bare part must sit entirely on the pad, without reaching the next one.
3. Tin the two stripped ends: a drop of solder on the bare copper.
4. Lay the wire on the copper side with its ends over the two pads. Hold it still with tweezers.
5. Heat the solder already on each pad: the end of the wire sinks into it. Add a little solder if needed.

> **Why insulated wire**
>
> Keeping the insulation and stripping only the ends avoids accidental contact between neighbouring columns, for example between column 2 (VSYS) and column 3 (ground). When two runs must meet in the same hole, use two pieces of wire with their ends soldered on the same pad.

3.4 Using the multimeter

Mode	Symbol	What it is for here	How to read it
Continuity	♪ or)))	Check that two holes are connected, or that they are NOT	Beep = connected. No beep = isolated.
Diode test	▶|	Check the direction of the diode and of the PC817 LED	Red probe on the anode, black on the cathode: it reads a voltage (0.15 to 1.2 V depending on the part). Reversed: OL.
DC voltage	V⎓ (20 V)	Measure 5V standby, VSYS and pin 9 of TPMS1	Black probe on ground, red on the point to measure.
Resistance	Ω	Identify the resistors if you are unsure about the colours	330 Ω and 1 kΩ (1000 Ω), within a few % tolerance.

> **Use the multimeter with the board unpowered**
>
> Continuity, diode test and resistance must be measured **with no power** and, for this project, **without the Pico in the headers**: the Pico would alter the readings.

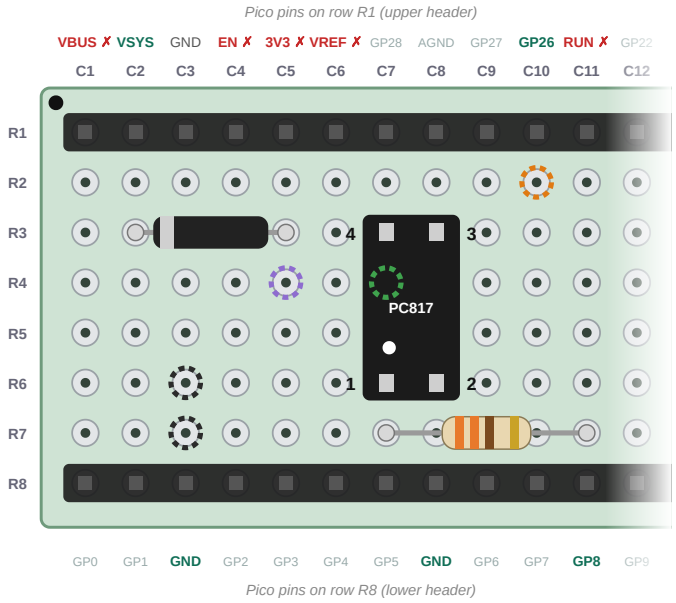
4 Step-by-step assembly

The steps go from the lowest components to the tallest, then the connections underneath, then the external wires. Keep this manual open on the current step and tick off the checklist in Appendix B as you go.

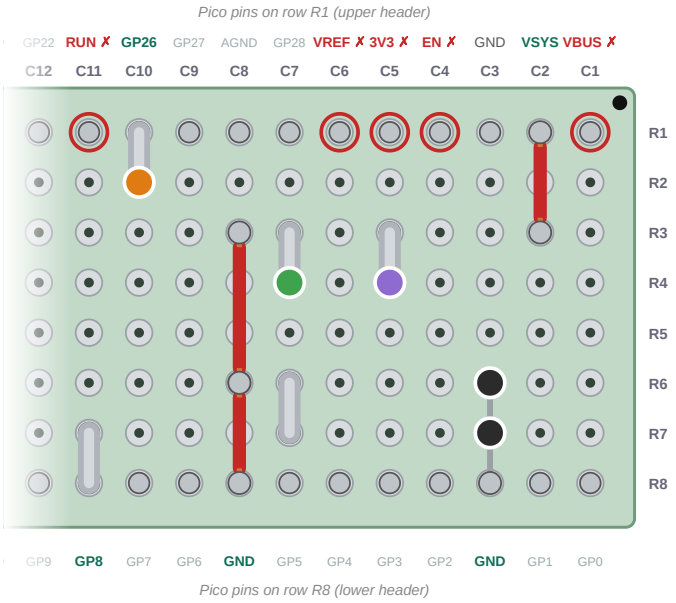
Step	What you do	Time
1	Cut the perfboard to 20 × 8 holes	15 min
2	1N5819 diode	10 min
3	330 Ω resistor	5 min
4	PC817 optocoupler	10 min
5	Female headers	20 min
6	Red wire on column 2 (VSYS)	10 min
7	Red wires on column 8 (ground)	10 min
8	Internal solder bridges	10 min
9	Preparing the external wires	20 min
10	Soldering the external wires and their bridges	20 min
11	Full check with the multimeter	15 min
12	Securing the wires and plugging in the Pico	10 min

Where you are heading

Component side · top



Copper side · bottom (mirrored)



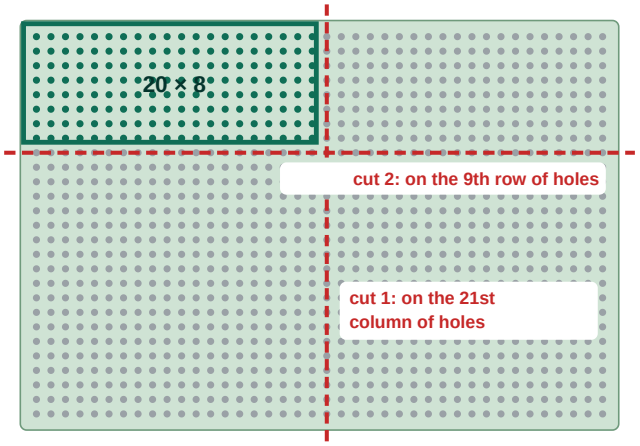
The finished board, columns 1–12 (13 to 20 hold only the headers). All connections are listed in Appendix A.

Before you start

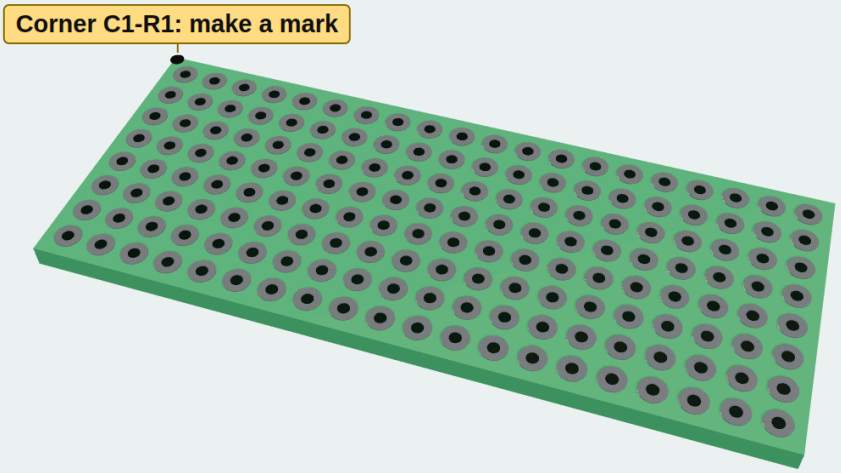
Good light and a magnifier · heat-resistant mat · multimeter in continuity mode within reach · the Pico **is not soldered**: it only goes into the headers at Step 12.

Step 1 Cut the perfboard

You need: the perfboard (yours is 40 × 27 holes), utility knife, metal ruler, file or sandpaper, marker · Time: 15 minutes



1. Pick a corner of the perfboard and count **20 columns** and **8 rows** of holes.
2. Lay the metal ruler on the line of holes right after them (the 21st column) and **score 6–8 times** with the knife along the holes, always in the same groove. Do the same on the other face of the board.
3. Put the board on the edge of the table with the score line on the edge and **snap it** with a sharp push.
4. Repeat for the long cut, on the 9th row of holes.
5. File the edges until smooth. About 1 mm of edge must remain beyond the last row of holes.
6. Hold the board with the **component side** towards you (the one without pads; if it has pads on both sides, pick one) and the long side horizontal: the hole at the **top left is C1-R1**. Put a black dot there with the marker.
7. Flip the board like the page of a book and mark the same hole on the copper side too: you will find it at the **top right**.



The cut perfboard seen from the component side: the mark is on corner C1-R1.

Fibreglass dust

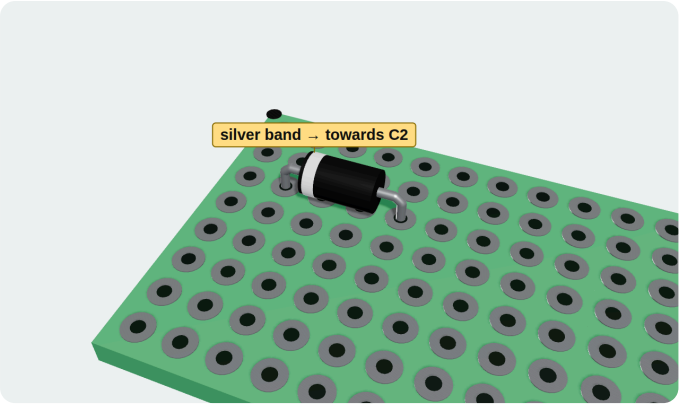
Scoring and snapping makes very little dust. If you use a saw, do it outdoors, with a mask, and clean up with a damp cloth.

Check

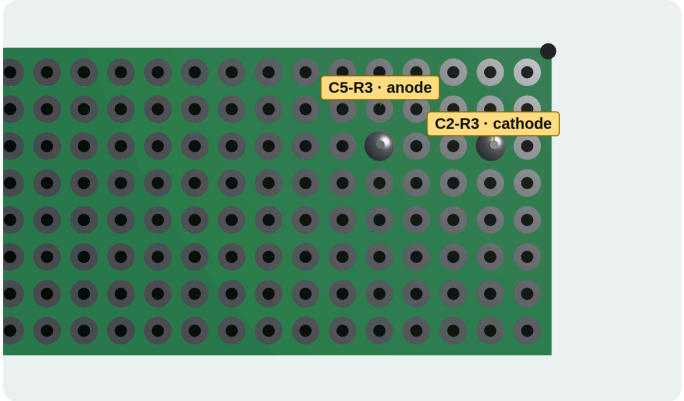
Count again: 20 holes on the long side and 8 on the short side. One row more or less shifts every connection.

Step 2 1N5819 diode

You need: 1N5819 diode, soldering iron, flush cutters · Time: 10 minutes



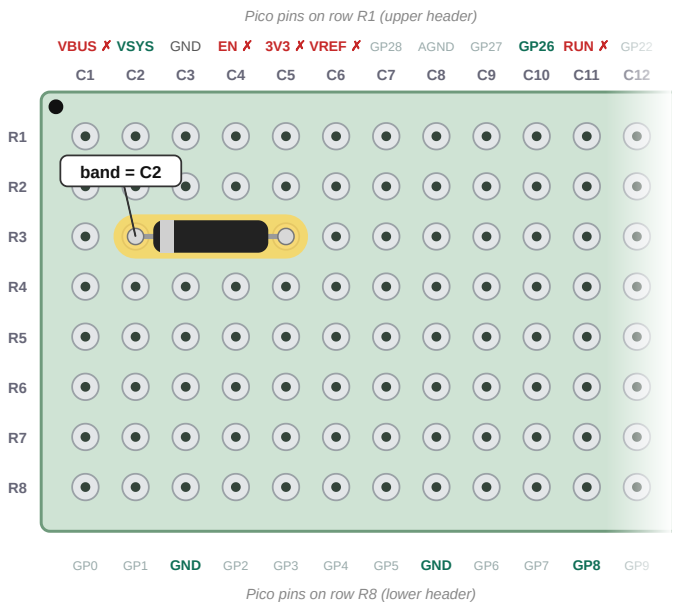
The diode with the silver band towards column 2.



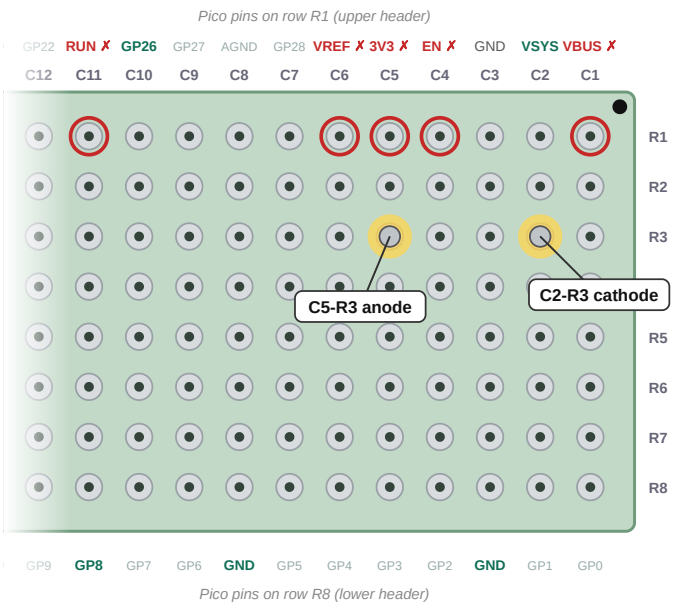
From the copper side (mirrored): the two joints.

1. Find the **silver band** on the diode body: that is the cathode.
2. Bend both leads at 90° **about 1 mm from the body**, holding the lead with tweezers resting against the body: this way the bend does not stress it. The leads must be 3 holes apart (7.6 mm): try them on the holes and tighten the bend if needed.
3. Insert the diode from the component side: **band in C2-R3**, the other lead in **C5-R3**. The body must rest on the board.
4. Flip the board and spread the leads slightly so the diode does not fall out.
5. Solder both leads (see 3.2) and trim them leaving 1–2 mm.
6. Keep the offcuts: you will need them for the bridges.

Component side · top



Copper side · bottom (mirrored)



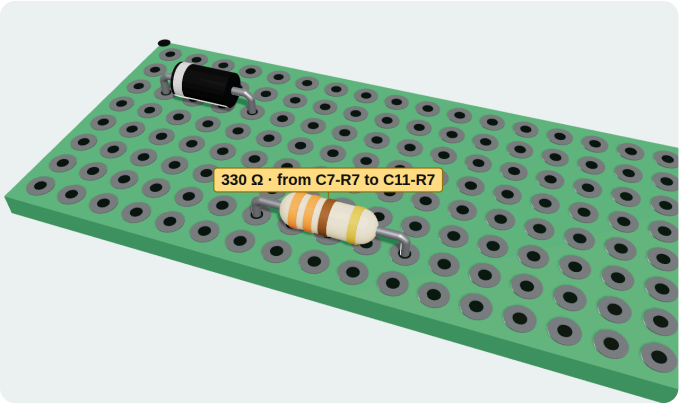
Detail of columns 1–12. The diode goes from C2-R3 (band) to C5-R3; below, the two joints.

Check with the multimeter in diode test

Red probe on C5-R3, black on C2-R3: **0.15–0.35 V**. Swapping the probes: **OL**. If you read the opposite, the diode is in backwards: desolder it and turn it round.

Step 3 330 Ω resistor

You need: 330 Ω resistor (orange-orange-brown-gold) · Time: 5 minutes



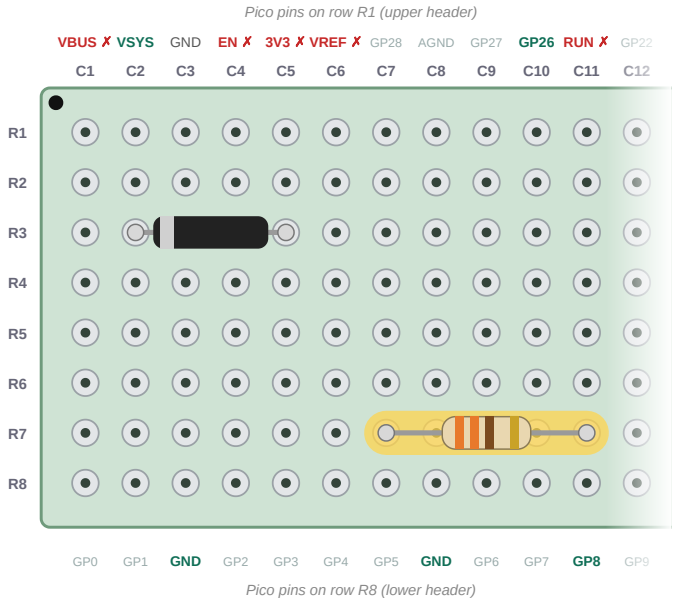
The resistor between C7-R7 and C11-R7.

1. Check the colours: **orange, orange, brown, gold**. If in doubt, measure it with the multimeter in Ω mode: about 330.
2. Bend the leads so they are 4 holes apart (10.2 mm).
3. Insert it in **C7-R7** and **C11-R7**, resting on the board. Resistors have no direction.
4. Flip, solder both leads and trim them to 1–2 mm.

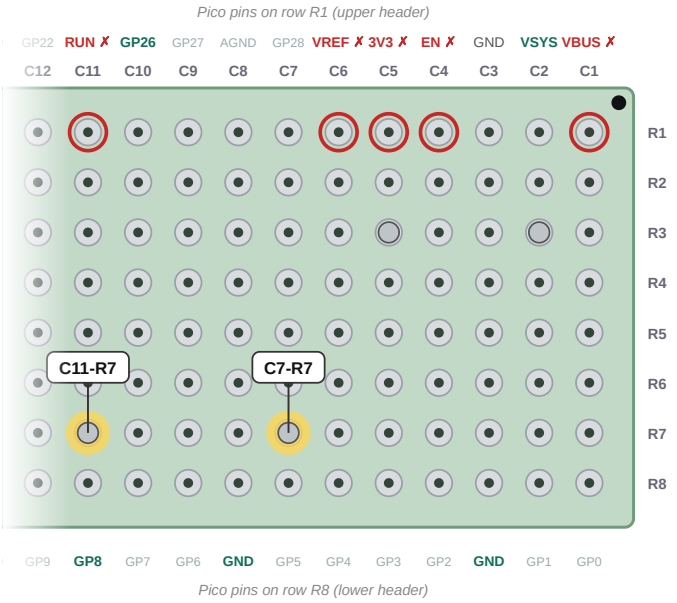
Tip

Bend the leads around a screwdriver or the tip of the cutters: the curve comes out even and the resistor sits straight.

Component side · top



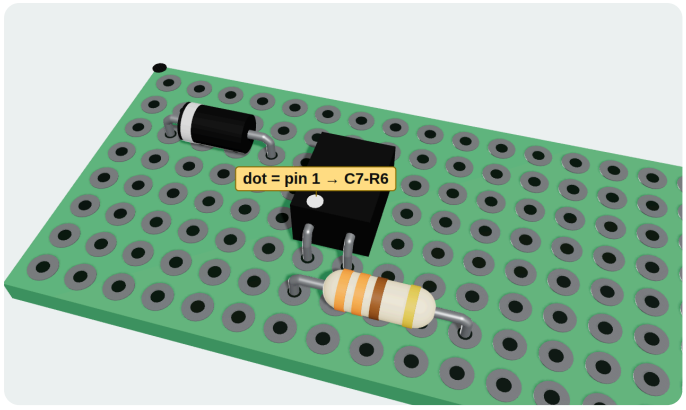
Copper side · bottom (mirrored)



Detail of columns 1–12: the resistor on row R7, from C7 to C11.

Step 4 PC817 optocoupler

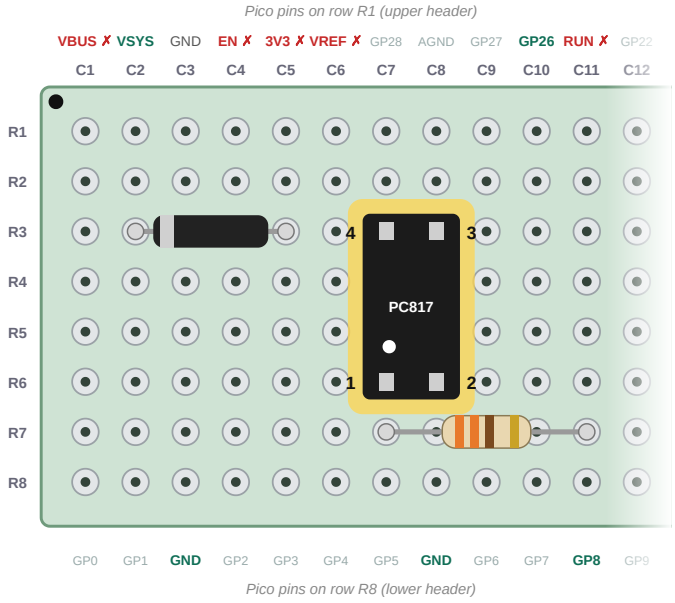
You need: PC817 (4-pin DIP) · Time: 10 minutes



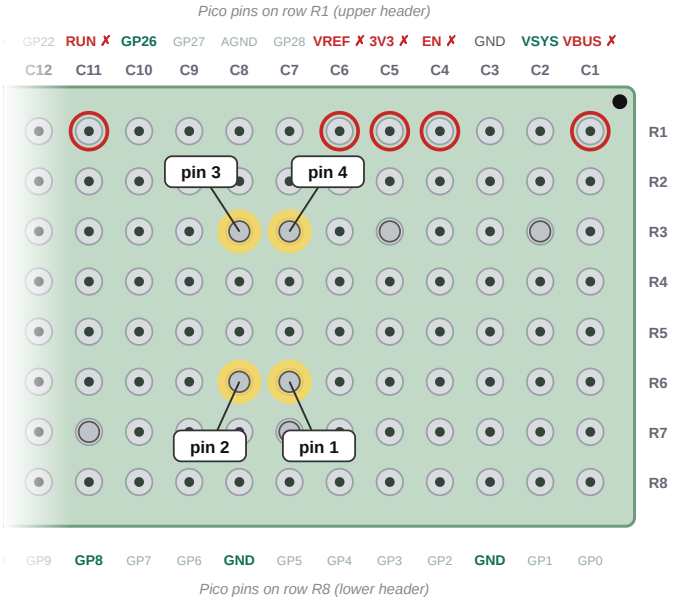
The PC817 with the dot (pin 1) at the bottom left, on hole C7-R6.

1. Look at the PC817 from above: the **dot** (or notch) marks **pin 1**. The others are counted anticlockwise: 1, 2, 3, 4.
2. If the pins are splayed too wide, lay the PC817 on its side on the table and press gently to straighten them.
3. Insert it with: **pin 1 in C7-R6**, pin 2 in C8-R6, pin 3 in C8-R3, pin 4 in C7-R3. The dot faces the 330 Ω resistor.
4. Flip the board and solder **two opposite pins** (for example 1 and 3). Check that the PC817 sits flat; if not, reheat and adjust.
5. Solder the other two pins. No need to trim them: they are already short.

Component side · top



Copper side · bottom (mirrored)



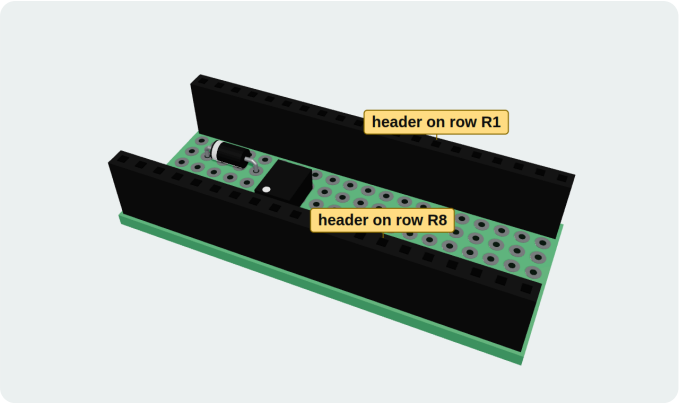
Detail of columns 1–12. The PC817 takes columns 7–8, rows 3–6. On the copper side, mirrored, pin 1 appears to the right of pin 2.

Check with the multimeter in diode test

Red probe on C7-R6 (pin 1), black on C8-R6 (pin 2): **about 1.0–1.2 V** (the internal LED). Reversed: OL. If not, the PC817 is rotated.

Step 5 Female headers

You need: 2 female 1 × 20 headers, the Pico 2 WH (as a jig) · Time: 20 minutes

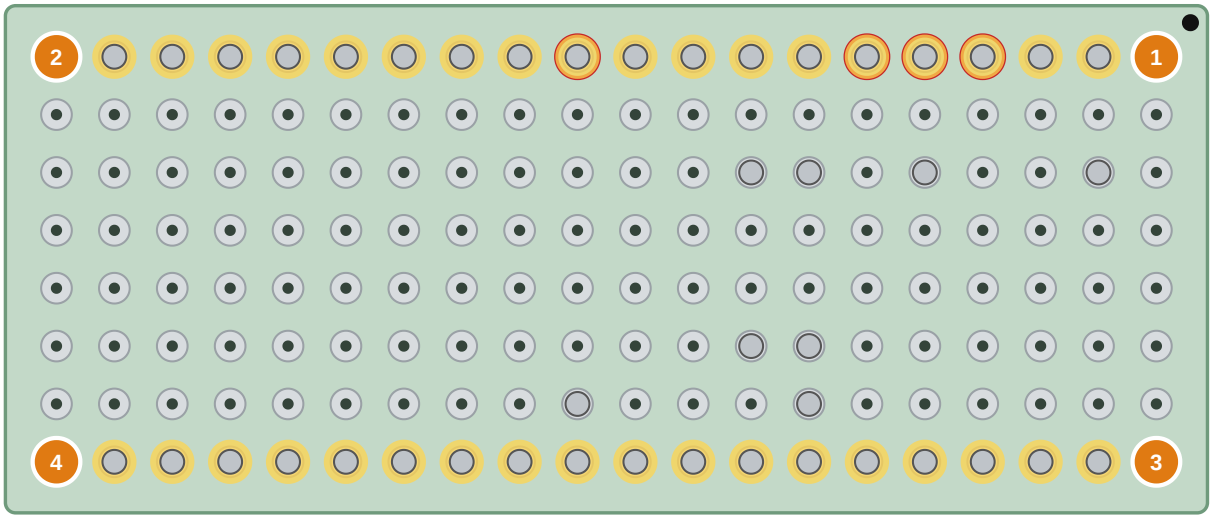


The two headers on rows R1 and R8.

1. Push the two headers onto the Pico pins, as if you were mounting it: this keeps them perfectly parallel and at the right spacing.
2. Rest the Pico + headers assembly on the **component side**: pins in rows **R1 and R8**, columns 1 to 20, with the **USB socket towards the column 1 edge** (the dot).
3. Flip everything carefully and solder **only the 4 corner pins**: C1-R1, C20-R1, C1-R8, C20-R8.
4. Check that the headers sit flat on the board. Then **remove the Pico**: hold the headers down on the board with two fingers and pull it out rocking gently, so you do not pull on the 4 joints.
5. Solder the other 36 pins, without the Pico: this way you do not heat it for nothing.

Pico pins on row R1 (upper header)

GP16	GP17	GND	GP18	GP19	GP20	GP21	GND	GP22	RUN ✗	GP26	GP27	AGND	GP28	VREF ✗	3V3 ✗	EN ✗	GND	VSYS	VBUS ✗
C20	C19	C18	C17	C16	C15	C14	C13	C12	C11	C10	C9	C8	C7	C6	C5	C4	C3	C2	C1



GP15	GP14	GND	GP13	GP12	GP11	GP10	GND	GP9	GP8	GP7	GP6	GND	GP5	GP4	GP3	GP2	GND	GP1	GP0

Pico pins on row R8 (lower header)

Copper side: 40 joints, two rows of 20. First the four numbered corners, then all the others.

Mind the neighbouring pins

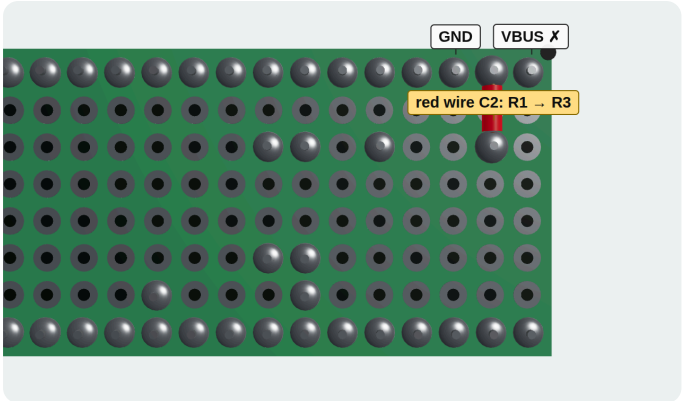
On row R1 the pins are close together: one blob too many can join VBUS (C1) to VSYS (C2), or VSYS to GND (C3). Use little solder and look at every joint.

Check

With the multimeter in continuity, between C1-R1 and C2-R1 it **must not beep**, nor between C2-R1 and C3-R1.

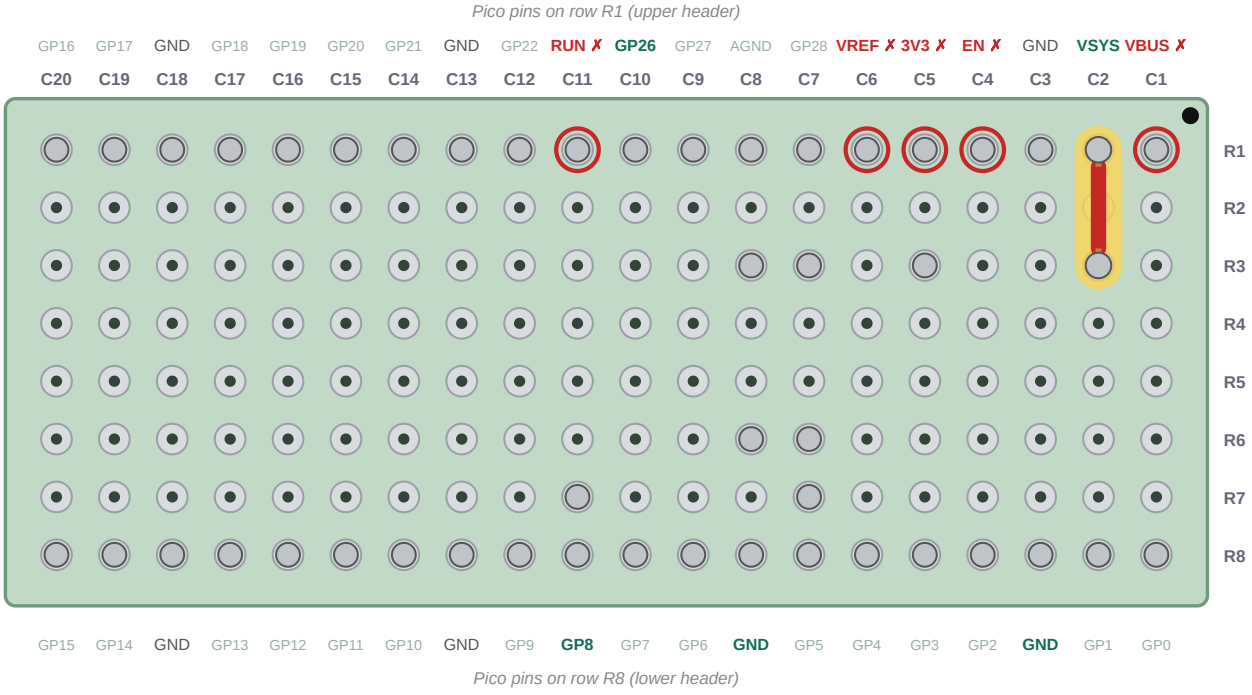
Step 6 Red wire on column 2 (VSYS)

You need: red solid-core wire, wire strippers · Time: 10 minutes



Copper side: the red wire joins the VSYS pin (C2-R1) to the diode cathode (C2-R3).

1. Cut a piece of red wire about 7 mm long (the 5 mm run plus 2 mm).
2. Strip 2 mm at each end and tin them.
3. Lay the wire on the copper side along **column 2**: one end on the joint of **C2-R1** (header, VSYS pin), the other on the joint of **C2-R3** (diode cathode).
4. Reheat the two joints until the ends sink into them.
5. Hole C2-R2, in the middle, stays under the insulation: do not solder it.



Copper side: the red run on column 2, from R1 to R3.

Warning: the most delicate area of the circuit

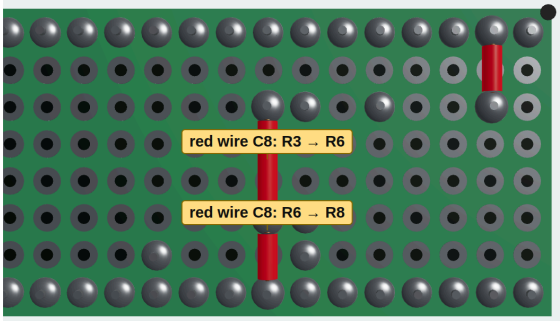
To the right of C2-R1 is **VBUS** (C1-R1), to the left **GND** (C3-R1). The solder of the red wire must touch neither: contact with GND would short the 5V standby rail.

Check

Continuity between C2-R1 and C2-R3: **must beep**. Between C2-R1 and C3-R1: **must not beep**.

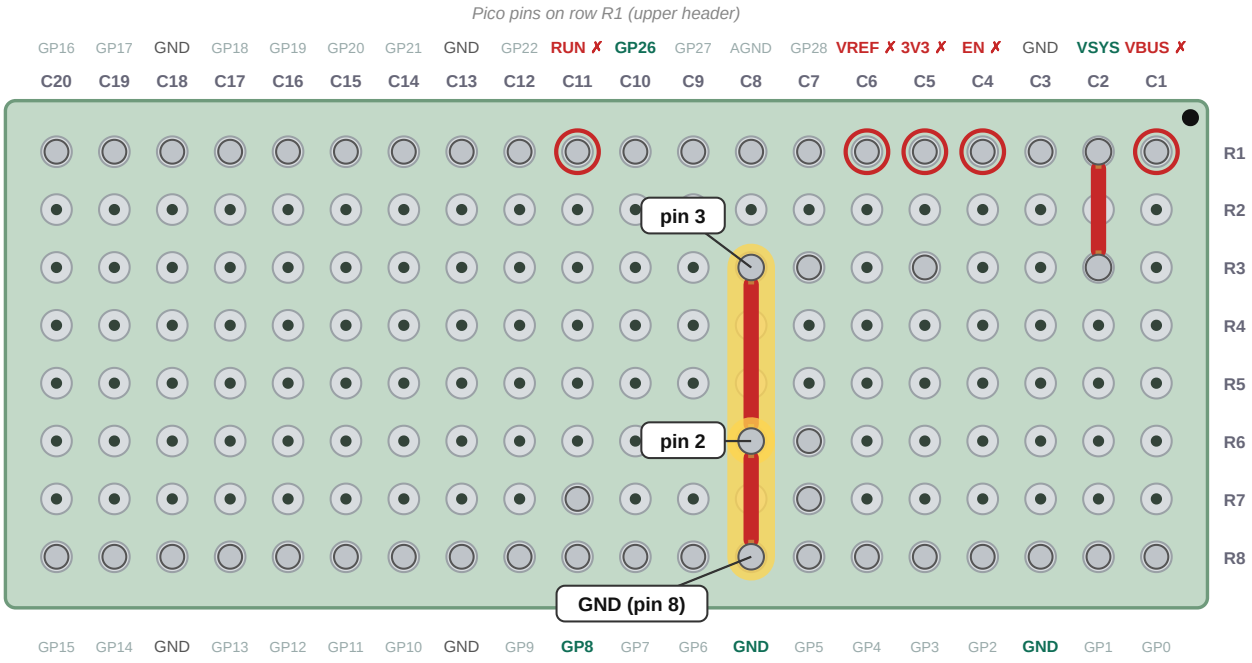
Step 7 Red wires on column 8 (ground)

You need: red solid-core wire · Time: 10 minutes



Copper side: two pieces of red wire along column 8.

1. Prepare **two pieces** of red wire, stripped 2 mm and tinned at the ends: one about 10 mm long (R3 to R6), one about 7 mm (R6 to R8).
2. First piece: from **C8-R3** (PC817 pin 3) to **C8-R6** (PC817 pin 2).
3. Second piece: from **C8-R6** to **C8-R8** (Pico GND pin 8, on the header).
4. Two ends meet at C8-R6: solder them together on the same pad, over the PC817 pin.
5. The upper part of column 8 (R1 and R2) stays free.



Copper side: column 8 takes PC817 pins 2 and 3 to ground.

Why both to ground

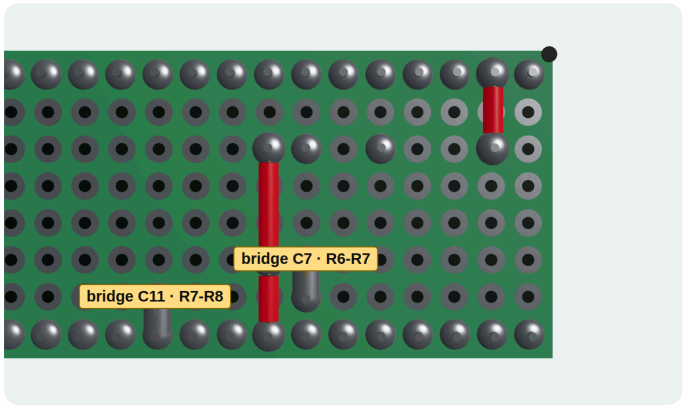
Pin 2 closes the LED circuit, pin 3 is the point the phototransistor pulls PS_ON down to. The Pico and the PSU share the same ground (the Pico is powered from 5V standby), so joining them is correct.

Check

Continuity between C8-R3 and C8-R8: **must beep**. Between C7-R3 (pin 4) and C8-R3 (pin 3): **must not beep**, otherwise the PSU would stay on all the time.

Step 8 Internal solder bridges

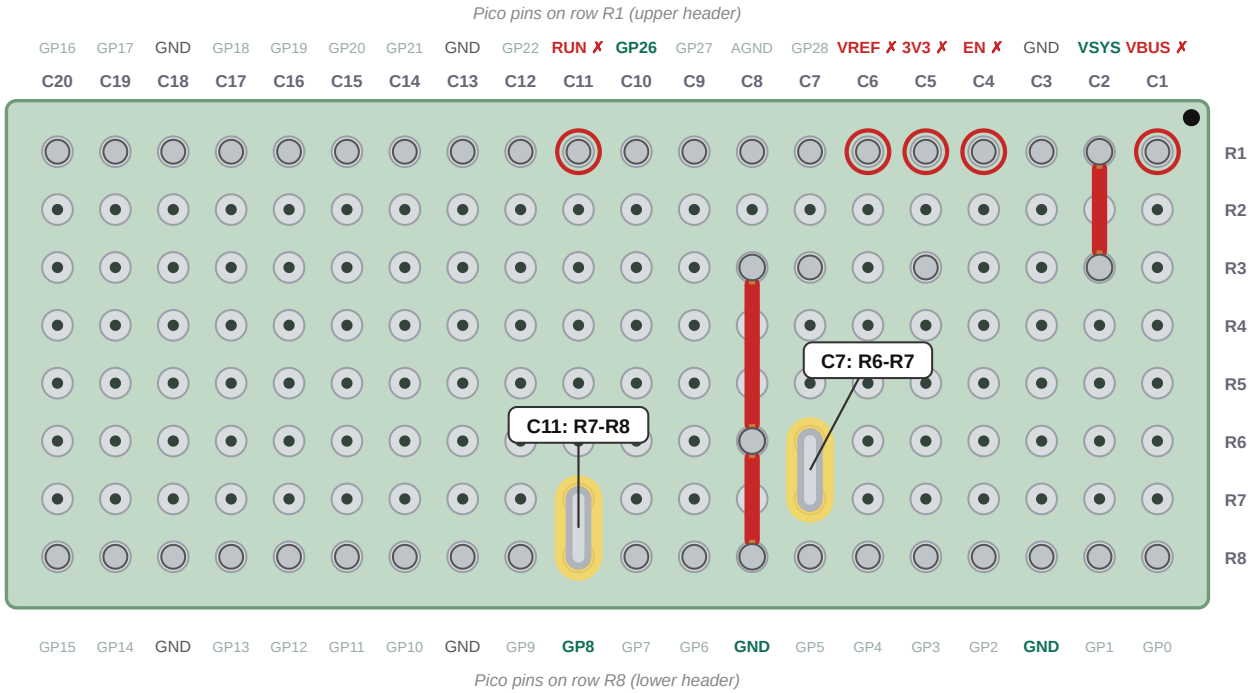
You need: solder, lead offcuts · Time: 10 minutes



Copper side: the two internal bridges.

These two bridges connect the 330 Ω resistor: on one side to the Pico's GP8 pin, on the other to PC817 pin 1.

1. **Bridge C11-R7 ↔ C11-R8**: joins the 330 Ω lead in C11-R7 to the GP8 header pin (C11-R8).
2. **Bridge C7-R6 ↔ C7-R7**: joins PC817 pin 1 (C7-R6) to the 330 Ω lead in C7-R7.
3. If the solder will not bridge by itself, lay a lead offcut between the two holes and solder it.



Copper side: the two bridges highlighted.

Do not widen the C7 bridge

The C7 R6–R7 bridge runs next to column 8, which is ground: it must not touch C8-R6 (pin 2) or the red wire. And hole **C7-R5 must stay empty**: it separates the LED (below) from PS_ON (above).

Check

Continuity C11-R7 ↔ C11-R8: beeps. C7-R6 ↔ C7-R7: beeps. C7-R6 ↔ C8-R6: does **not** beep (the LED would be shorted). C7-R7 ↔ C8-R8: does **not** beep.

Step 9 Prepare the external wires

You need: ATX extension (or breakout), 2 jumper leads 2.0 mm, 1 kΩ resistor, heat-shrink · Time: 20 minutes

From the power supply: purple, black, green

1. From the ATX extension keep three wires: **purple** (pin 9, 5V standby), **black** (pin 15 or 17, ground), **green** (pin 16, PS_ON).
2. The other wires of the extension must be **removed from the connector** or cut short and insulated one by one with heat-shrink: with the PSU on they carry 12 V, 5 V and 3.3 V.
3. Length: enough to reach where the board will sit in the case, plus 5 cm.
4. Strip 3 mm and tin the end of each one.

Identify the ATX wires with the multimeter

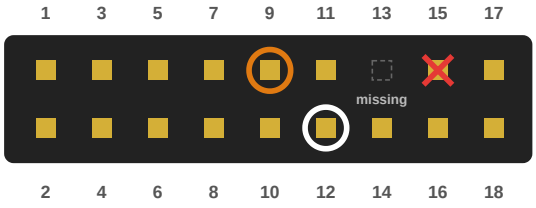
PSU in standby (plugged in, fans still), black probe on black: **purple** reads about **5 V**, **green** between 3 and 5 V, grey (unused) 0 V.

From the BC250: orange and black

1. Take the two leads with the 2.0 mm female end: the **orange** one goes on TPMS1 pin 9, the **black** one on pin 12 (figure below).
2. Cut off the other end, strip 3 mm and tin.
3. On the orange wire, 5–8 cm from the board, solder the **1 kΩ resistor in series** (brown-black-red) and cover it with heat-shrink (last figure).

Labels

Put a piece of tape on each wire saying what it is for (5VSB, GND, PS_ON, SENS, GND2): in the case there are two black wires and they get mixed up.



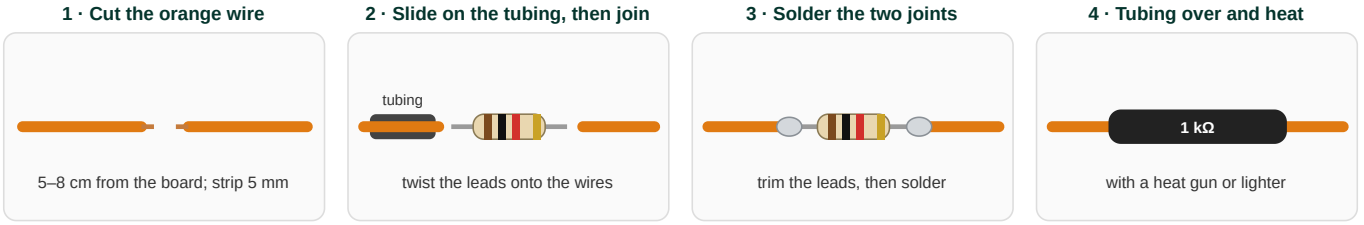
pin 9 = 3V: present only with the BC250 on → orange wire

pin 12 = ground → black wire

pin 15 = 3VSB: stays on even with the BC250 off. Do NOT use it

How to find them: start from the empty position of pin 13.
On the side with more pins: the 1st is 11, the 2nd is 9. On the other side: 15.
Pin 12 is in the other row, opposite 11. Pitch 2.0 mm.

The BC250 TPMS1 header: 18 positions in two rows, 2.0 mm pitch, pin 13 missing. Identify the pins by position; in Test 2 you will confirm them with the multimeter.



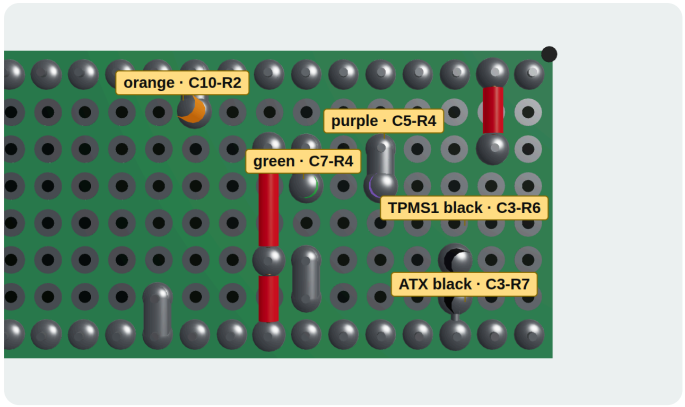
The 1 kΩ resistor in series with the orange wire, protected by heat-shrink.

If the ATX wires are too thick

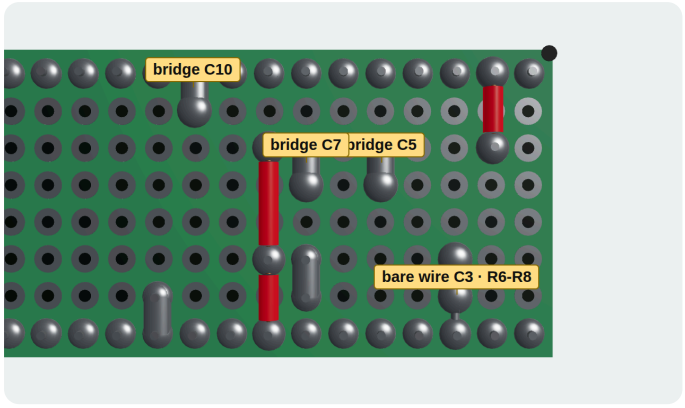
ATX extension wires are often 18 AWG: once tinned they do not fit the 1 mm holes. Do not enlarge the holes: solder 3 cm of solid-core wire to the tip of each one, insulate the splice with heat-shrink and use that thin pigtail for the hole (the label tells you which wire it is).

Step 10 Solder the external wires and their bridges

You need: the 5 prepared wires, lead offcuts · Time: 20 minutes



Copper side: where the five external wires go in.



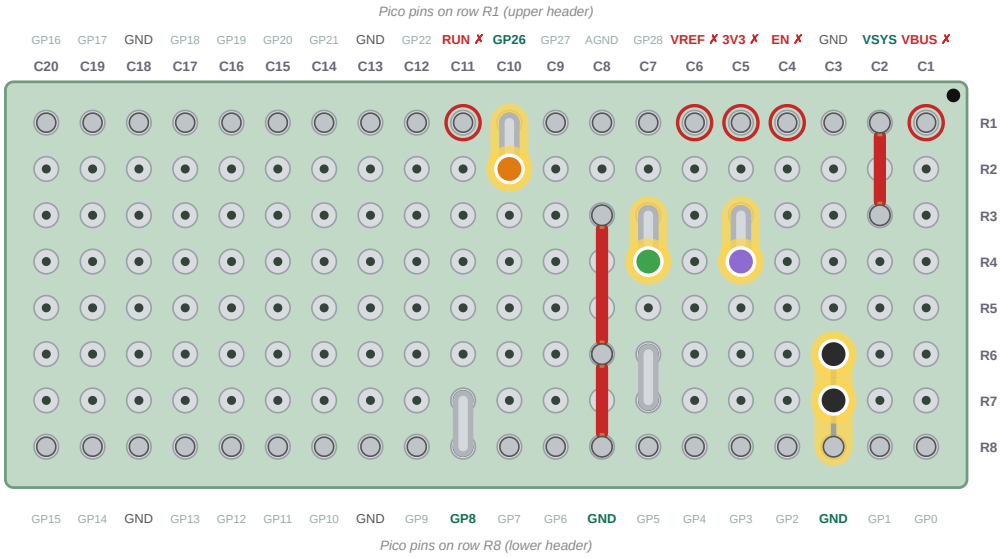
Copper side: the bridges of the external wires and the bare wire on column 3.

The external wires go in **from the copper side**: the Pico is above, and the wires must leave from below.

1. Push the stripped tip of each wire into its hole **from the copper side**, about 1 mm (less than the board thickness): it must not come out the other side, where the PC817 and the Pico are. Solder it on the copper side.
2. **Purple** (5VSB) in **C5-R4**, then make the **bridge C5-R4 ↔ C5-R3** (diode anode).
3. **Green** (PS_ON) in **C7-R4**, then the **bridge C7-R4 ↔ C7-R3** (PC817 pin 4).
4. **Orange** (sensor, with the 1 kΩ) in **C10-R2**, then the **bridge C10-R2 ↔ C10-R1** (GP26 pin).
5. **TPMS1 black** in **C3-R6** and **ATX black** in **C3-R7**. Then lay a lead offcut along column 3 from R6 to R8 and solder it at the three points: it connects the two blacks to GND pin 3 (C3-R8).
6. Look at the component side: if any tip sticks out, trim it flush. In C7-R4 the tip must not stick out at all: the PC817 is above it and you cannot trim there.

Keep 5V standby and PS_ON apart

C5 (purple) and C7 (green) are close: between them is column 6, which must stay empty.



Copper side: all connections done. Coloured circles = external wires.

The C10 bridge is next to RUN

Next to C10-R1 (GP26) is **C11-R1 = RUN**. If solder joins them, with the BC250 off the Pico stays stuck in reset and can no longer turn it on.

Step 11 Full check with the multimeter

You need: multimeter · Time: 15 minutes · Pico NOT inserted, PSU unplugged

1. Set the multimeter to **continuity**.
2. Check every pair in the two tables: the green ones (letters) must beep, the red ones (numbers) must not.
3. Switch to **diode test** and repeat the two checks on D1 and the PC817.
4. If a red pair beeps, look for the excess solder between those two holes with a magnifier and remove it with desoldering braid or by reheating.
5. Trick: photograph the copper side with your phone and zoom in. Excess blobs show up better in a photo than by eye.

Final diode test

D1: red on C5-R3, black on C2-R3 → 0.15–0.35 V.
PC817: red on C7-R6, black on C8-R6 → about 1.1 V.
Swapping the probes: OL.

External wires, end to end

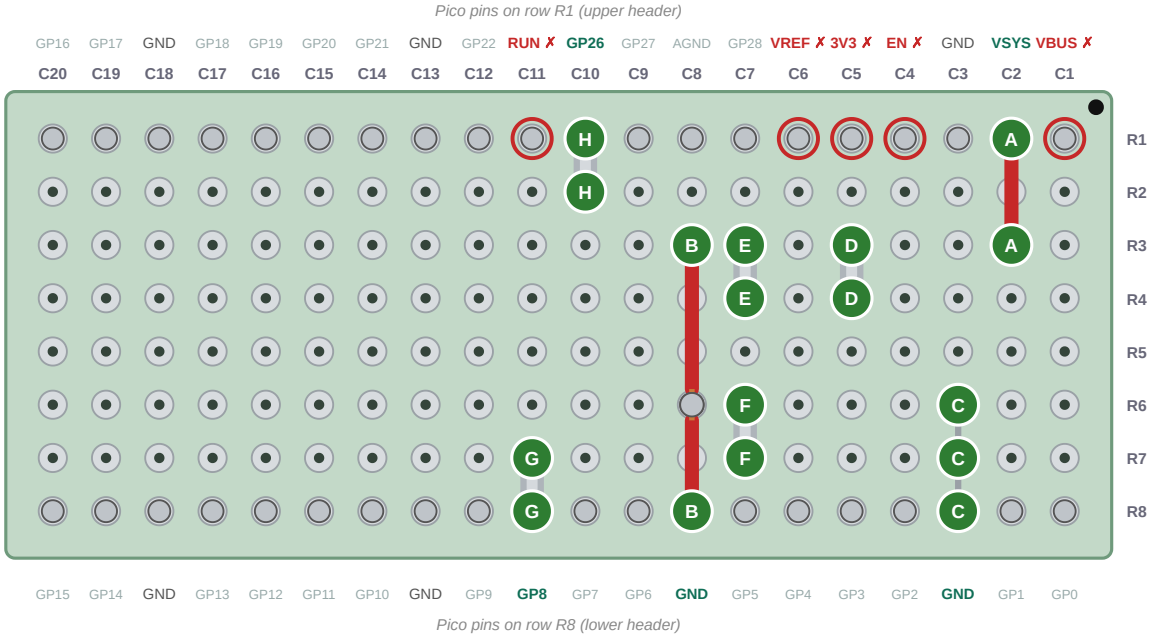
In continuity: ATX connector pin 9 ↔ C5-R3 · pin 16 ↔ C7-R3 · pin 15 or 17 ↔ C3-R8 · black female ↔ C3-R8. In Ω mode: orange female ↔ C10-R1 ≈ 1 kΩ. This catches two swapped wires right away.

Must beep

	Between	and
A	C2-R1 (VSYS)	C2-R3
B	C8-R3	C8-R8
C	C3-R6 and C3-R7	C3-R8
D	C5-R3	C5-R4
E	C7-R3	C7-R4
F	C7-R6	C7-R7
G	C11-R7	C11-R8
H	C10-R1	C10-R2

Must not beep

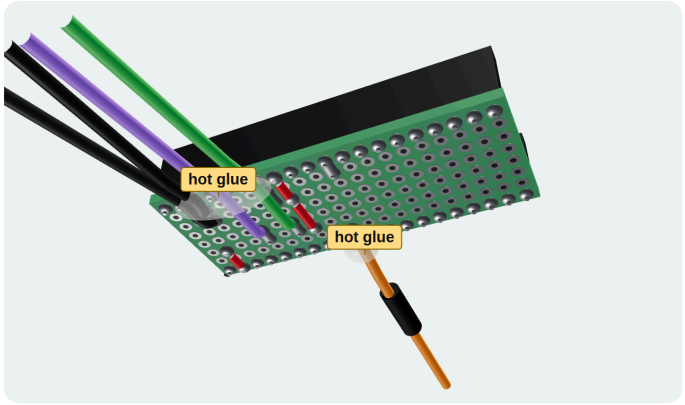
	Between	and
1	C1-R1 (VBUS)	C2-R1 (VSYS)
2	C2-R1 (VSYS)	C3-R1 (GND)
3	C4-R1 (EN)	C3-R1 (GND)
4	C11-R1 (RUN)	C10-R1 (GP26)
5	C5-R4 (5VSB)	C3-R8 (GND)
6	C7-R3 (PS_ON)	C8-R3 (GND)
7	C7-R4 (PS_ON)	C7-R6 (LED)
8	C5-R4 (5VSB)	C7-R4 (PS_ON)



Copper side: the points that must beep with each other (A–H). The pairs that must not beep are in the red table.

Step 12 Secure the wires and plug in the Pico

You need: hot glue or zip ties, the Pico 2 WH · Time: 10 minutes



Hot glue on the wires, near the edge of the board.

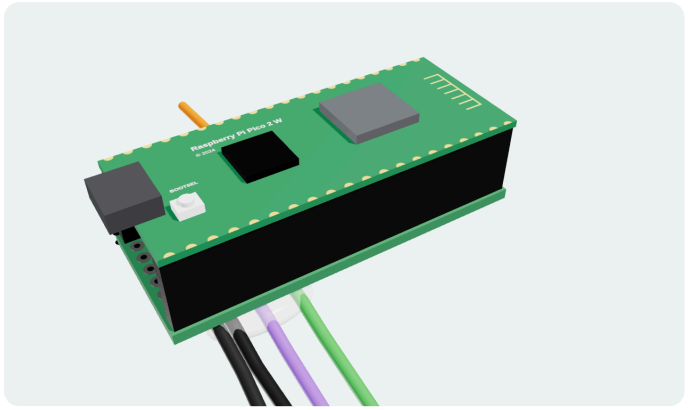
1. Put a drop of **hot glue** on the wires, on the copper side, near where they leave the board. If they get pulled, the glue takes the force and not the solder joint.
2. Wait for the glue to cool (1 minute).
3. Take the Pico 2 WH: **USB socket towards the column 1 edge** (the dot), pins **GP0…GP15 towards row R8**, the VBUS/VSYS pins towards row R1.
4. Line the pins up with the headers and push with both thumbs, evenly, until the Pico is fully seated.

Pico orientation

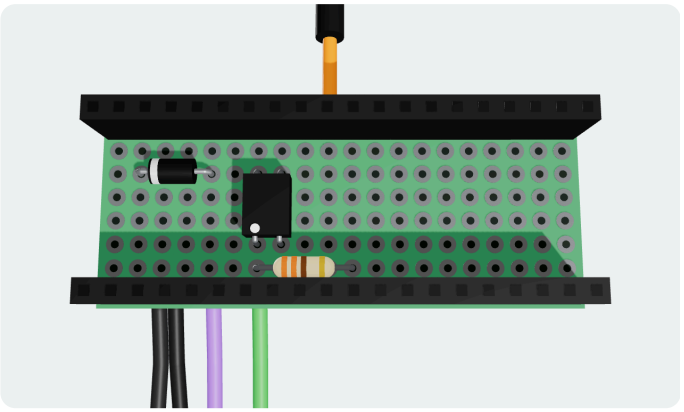
If the Pico is rotated 180°, the USB ends up on the column 20 side: remove it at once, before applying power.

Check

The Pico is straight, at the same height on both headers, and no wire is trapped under it.



The finished board, ready for testing.



What it looks like under the Pico, for reference.

5 Tests

Run the tests in order. For all three you will use **MicroPython**, which installs in a minute and needs no compiling. You will load the final firmware (chapter 7) afterwards.

Preparation: MicroPython on the Pico

1. Download the MicroPython `.uf2` file for the **Raspberry Pi Pico 2 W** from micropython.org.
2. With the PSU **unplugged from the wall**, hold down the Pico's **BOOTSEL** button and connect it to the PC with the micro-USB cable: it shows up as a USB drive. If the Pico is already getting 5V standby, holding BOOTSEL while plugging the cable has no effect.
3. Drag the `.uf2` file onto the drive: the Pico reboots into MicroPython.
4. Install **Thonny** on the PC, open *Tools* → *Options* → *Interpreter* and choose *MicroPython (Raspberry Pi Pico)*.

> **The Pico can stay on the board**
>
> Thanks to diode D1, the Pico can be connected to the PC over USB and to 5V standby at the same time.

Test 1 · Power supply and Pico (no BC250)

1. Leave the BC250's 8-pin **PCIe cable unplugged**, unplug the TPMS1 leads and also the **Pico's USB cable**.
2. Connect purple, black and green to the PSU (breakout or extension). **Remove any jumper** between green and black that you were using until now.
3. Plug in and flip the switch: the fans must stay **still**.
4. With the multimeter on voltage, black on C13-R8 (a header GND) and red on C2-R3 (D1 cathode, i.e. VSYS): about **4.7 V**. With USB disconnected, this reading proves that 5V standby reaches the Pico.
5. Connect the Pico to the PC, open Thonny and run:

```
from machine import Pin
import time
ps_on = Pin(8, Pin.OUT, value=0)
ps_on.value(1)          # the PSU must start (fan)
print("PSU on")
time.sleep(10)
ps_on.value(0)          # it must go back to standby
print("PSU in standby")
```

> **Expected result**
>
> The PSU runs for 10 seconds and then goes back to standby. If it has a semi-passive fan that does not spin with no load, measure 12 V on the PCIe connector during those 10 seconds (yellow wire against black). If it does not start: go over checks B, E, F, G and 8 of step 11 and the PC817 diode test.

Test 2 · The BC250 sensor

1. Connect the PCIe cable to the BC250.
2. Locate the TPMS1 pins **by position** (figure in Step 9): find the empty position of pin 13; in the same row, on the side with more pins, the first is 11 and the second is **9**. Right after the gap is 15; **12** is in the other row, opposite 11.
3. In the Thonny shell type `from machine import Pin`, press Enter, then `Pin(8, Pin.OUT, value=1)`: the PSU turns on and the BC250 boots.
4. Confirm with the multimeter, black on a ground: pin 9 and pin 15 give about **3.3 V**. The pins are tightly packed: place the probe calmly, without touching two at once.
5. Shut down from SteamOS leaving the PSU on: **pin 9 drops to 0 V**, pin 15 stays at 3.3 V.
6. Type `Pin(8, Pin.OUT, value=0)` to turn the PSU off and unplug it. With the multimeter in continuity check that pin 12 beeps with the PSU black wire, then fit the **orange lead on pin 9** and the **black one on pin 12** and plug back in.
7. Run the script below: it turns the PSU on and prints the BC250 state every second.

```
from machine import Pin
import time
ps_on = Pin(8, Pin.OUT, value=1)          # turns the PSU on
sensor = Pin(26, Pin.IN, Pin.PULL_DOWN)
while True:
    print("BC250 ON" if sensor.value() else "BC250 off")
    time.sleep(1)
```

Expected result

A few seconds after power-on you read "BC250 ON". After shutting down from SteamOS, a few seconds later you read "BC250 off". Only now press Stop in Thonny.

Sudden power-off

If the Pico reboots (also when you press Stop or Restart in Thonny) GP8 can go low: the PSU cuts out at once together with the BC250. With the BC250 on, do not stop the scripts: shut it down from SteamOS first. Do not keep work open during the tests.

Test 3 · Full cycle with the BOOTSEL button

This script already does the whole power-control job, using the BOOTSEL button in place of the controller. Start it with the BC250 off.

```
from machine import Pin
import rp2, time
sensor = Pin(26, Pin.IN, Pin.PULL_DOWN)
s = sensor.value()
ps_on = Pin(8, Pin.OUT, value=s)    # safe start: if the BC250 is on it stays on
state, t = ("ON" if s else "OFF"), time.ticks_ms()
print("Press BOOTSEL to power on")
while True:
    now, s = time.ticks_ms(), sensor.value()
    if state == "OFF" and rp2.bootsel_button():
        ps_on.value(1); state, t = "STARTING", now; print("Powering on")
    elif state == "STARTING":
        if s: state = "ON"; print("BC250 on")
        elif time.ticks_diff(now, t) > 10000:
            ps_on.value(0); state = "OFF"; print("It did not start: powering off")
    elif state == "ON" and not s:
        state, t = "SHUTTING_DOWN", now
    elif state == "SHUTTING_DOWN":
        if s: state = "ON"
        elif time.ticks_diff(now, t) > 3000:
            ps_on.value(0); state = "OFF"; print("Off: standby")
    time.sleep_ms(100)
```

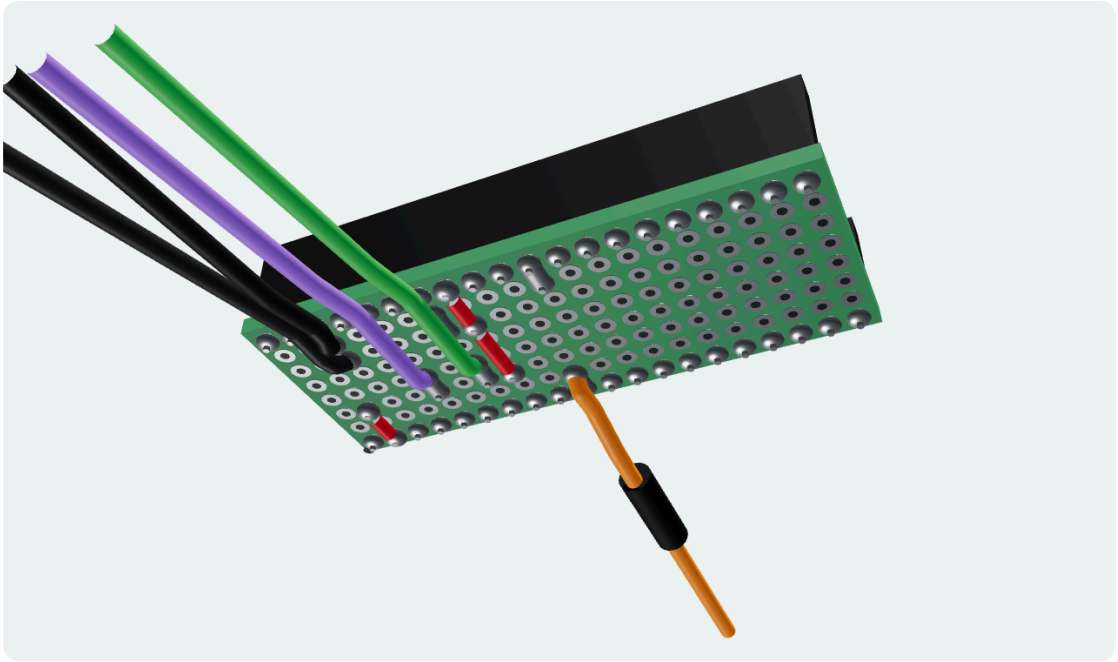
Expected result

Press BOOTSEL: the PSU starts and the BC250 boots. Shut down from SteamOS: after about 3 seconds the PSU goes to standby.

Stopgap solution

Saved on the Pico as `main.py`, the script starts by itself when you apply power. Until you have the final firmware, you can use the board like this, powering on with the BOOTSEL button.

6 Installing in the case



The wires leave from the copper side: you need room under the board.

1. **Do not rest the board directly on metal.** The copper side has exposed solder joints: use a 3D-printed holder that grips it by the edges, or standoffs at least 10–15 mm tall (the wires leave from the copper side), with Kapton tape underneath.
2. **Clear antenna:** the Pico's antenna is on the short edge opposite the USB (columns 17–20). Keep it away from sheet metal, from the BC250 and from USB 3 hubs; if the case is metal, point it towards a plastic part or outside the case.
3. **USB cable:** connect the Pico to a USB port of the BC250. Leave the cable slack, without pulling on the connector.
4. **TPMS1:** fit the 2.0 mm females on pins 9 (orange) and 12 (black). A drop of hot glue on the connector stops them working loose with vibration.
5. **ATX:** connect purple, black and green. Make sure you removed the old PS_ON jumper: if it stays, the PSU never turns off.
6. **BOOTSEL within reach:** if you can, leave a hole in the case above the button: you need it to pair controllers and to update the firmware (chapter 7).

> **Suspend**
> Disable suspend in SteamOS: in suspend, TPMS1 pin 9 could drop and the Pico would cut power to a suspended machine.

7 Final firmware

The final firmware combines the **DS5Dongle** dongle (github.com/awalol/DS5Dongle, MIT licence) with the power control from the tests. The logic is that of Test 3, safe start included, plus the 15 s pause after shutdown; power-on is triggered when a paired DualSense connects.

Parameter	Value
PS_ON output	GP8 (pin 11): high = PSU on. Recommended drive strength 8 mA
Sensor input	GP26 (pin 31) with internal pull-down enabled
Sensor polling	every 100 ms
Power-on timeout	10 s: if the BC250 does not start, release GP8
Shutdown delay	3 s of sensor low
Pause after shutdown	15 s during which controller connections are ignored
Safe start	at boot, if the sensor is already high, GP8 goes high at once and the logic starts in the ON state

The BOOTSEL button with DS5Dongle

Press	What it does
once	scans for controllers for 30 seconds (pairing)
twice	reboots the Pico
3 times	update mode (the USB drive for the .uf2 file appears)
long, 1.5 s	clears all pairings

Pairing a DualSense: hold Create and PS until the light bar blinks fast, then press BOOTSEL once on the Pico.

With the BC250 on, press BOOTSEL only once

Two or three presses reboot the Pico: GP8 drops and the PSU cuts out at once together with the BC250.

Updating the firmware

Always do it with the BC250 off. Safe method: unplug the PSU from the wall, disconnect the USB cable from the BC250 and connect it to the PC while holding BOOTSEL, then copy the new `.uf2`. With DS5Dongle three presses of BOOTSEL also work.

Prompt for Claude Code

```
Fork github.com/awalol/DS5Dongle for the Raspberry Pi Pico 2 W and add a power
controller for the ATX power supply of an AMD BC250 board.
- GP8 (output, 8 mA drive) drives the LED of a PC817 optocoupler that pulls PS_ON to
  ground: level signal, GP8 high = PSU on.
- GP26 (input, internal pull-down) reads pin 9 of the TPMS1 header through 1 kOhm:
  high = BC250 on.
- With the machine off, when a paired DualSense connects, drive GP8 high. If GP26
  does not rise within 10 s, release GP8.
- With the machine on, read GP26 every 100 ms; if it stays low for 3 consecutive s,
  drive GP8 low and ignore controller connections for the next 15 s.
- At boot, if GP26 is already high, drive GP8 high at once and start in the ON state.
- All the logic on core 1, watchdog enabled, no blocking waits on core 0.
- Do not change the existing Bluetooth, USB, audio, LED and BOOTSEL handling.
- First a minimal power-only version to test, then the integration with DS5Dongle.
```

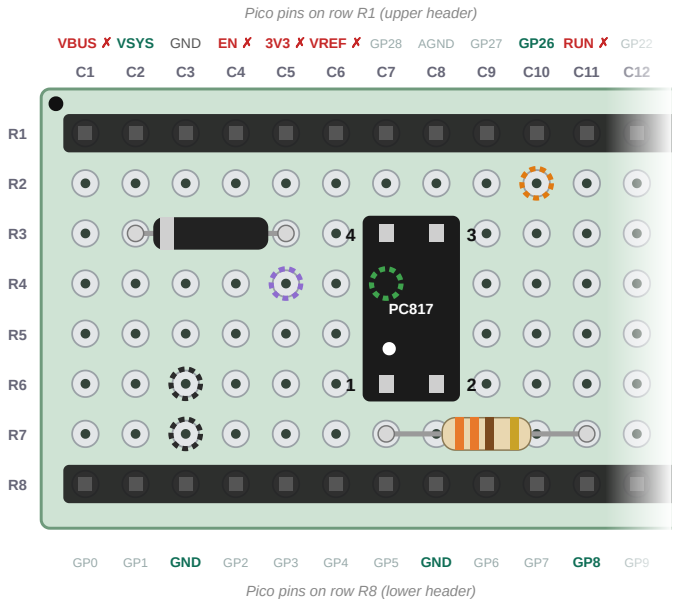
8 Troubleshooting

Symptom	Likely cause	What to do
With the machine off the Pico does not power up	Diode backwards; red wire C2, bridge C5 or ATX black not connected	Diode test on D1; checks A, C and D of step 11; measure 5 V on purple
With the Pico connected to the PC everything works, without the PC it does not	Same cause: the Pico is powered only from USB	As above: 5V standby does not reach VSYS
GP8 high but the PSU does not start	PC817 rotated; C7 or C11 bridges missing; green not on pin 16; solder between C5 and C7	Diode test on the PC817; checks B, E, F, G and 8 of step 11; recheck the wire colour
The PSU stays on all the time	Old jumper still fitted, or solder between C7-R3 and C8-R3	Remove the jumper; check 6 of step 11
The PSU starts and turns off after 10 s	The sensor signal does not arrive: orange on the wrong pin, C10 bridge missing	Test 2; check H; recheck pin 9
The sensor always says "on"	Orange on pin 15 (3VSB) instead of 9	Repeat the identification in Test 2 (points 2–5)
The sensor stays "on" with the BC250 off, but pin 9 measures 0 V	GP26 internal pull-down too weak: known defect of early RP2350 chips (stepping A2, erratum E9)	Solder a 4.7 kΩ resistor between the orange and black wires, at the TPMS1 connector end, and cover it with heat-shrink
The Pico does not boot or keeps rebooting	Solder between C10-R1 and C11-R1 (RUN) or between C3 and C4 on R1 (EN)	Checks 3 and 4 of step 11
The machine turns off while you play	Sensor wire working loose; poor contact on green or on bridges C7 and C11; Pico rebooted (BOOTSEL pressed 2 or 3 times)	Hot glue on the connectors; recheck the solder joints; with the BC250 on press BOOTSEL only once
The controller disconnects or lags	Pico antenna shielded	Move the board: antenna side towards plastic or outside the case
After shutdown it powers back on by itself	The controller stayed on and reconnects	15 s pause in the firmware; turn the controller off by holding PS for about 10 s

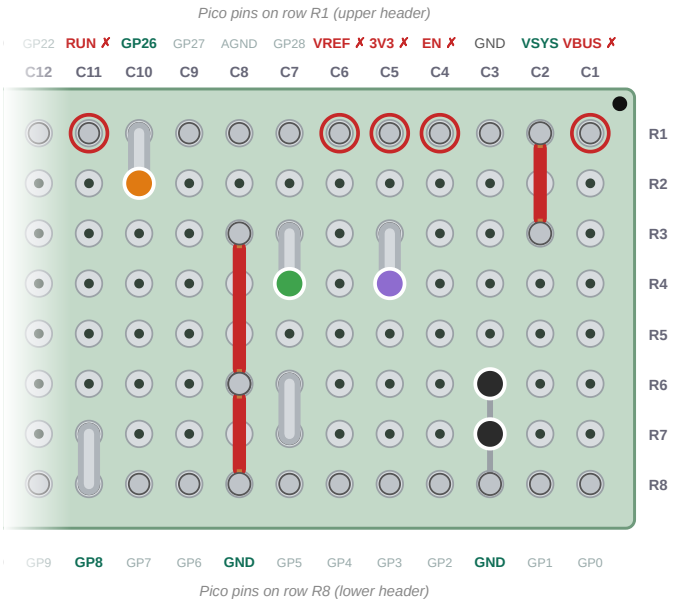
A Full connection table

Connection	Path
5V standby	purple wire → C5-R4 → bridge → C5-R3 (D1 anode)
VSYS	C2-R3 (D1 cathode) → red wire → C2-R1 (pin 39)
PS_ON	green wire → C7-R4 → bridge → C7-R3 (PC817 pin 4)
LED drive	C11-R8 (GP8, pin 11) → bridge → C11-R7 → R1 330 Ω → C7-R7 → bridge → C7-R6 (PC817 pin 1)
PC817 ground	C8-R3 (pin 3) → red wire → C8-R6 (pin 2) → red wire → C8-R8 (GND, pin 8)
Black wires ground	C3-R6 (TPMS1 black) · C3-R7 (ATX black) → bare wire → C3-R8 (GND, pin 3)
Sensor	TPMS1 pin 9 → orange wire with R2 1 kΩ → C10-R2 → bridge → C10-R1 (GP26, pin 31)
Holes to leave empty	All those not listed above. Watch especially: C2-R2 (under the red wire) · column 6 from R2 to R7 (separates 5V standby and PS_ON) · C7-R5 (separates the LED from PS_ON) · C8-R2 (AGND is above it) · C10 from R3 to R7.

Component side · top



Copper side · bottom (mirrored)



Complete assembly, columns 1–12 (13 to 20 hold only the header joints). On the component side the dashed circles are the wires coming in from below.

B Printable checklist

✓	Step	Done when…
☐	1 · Cut	20 × 8 holes counted, smooth edges, dot on C1-R1 on both sides
☐	2 · Diode	band in C2-R3; diode test 0.15–0.35 V
☐	3 · 330 Ω	C7-R7 → C11-R7, leads trimmed
☐	4 · PC817	pin 1 in C7-R6; diode test about 1.1 V
☐	5 · Headers	40 joints; VBUS–VSYS and VSYS–GND do not beep
☐	6 · C2 wire	C2-R1 ↔ C2-R3 beeps
☐	7 · C8 wires	C8-R3 ↔ C8-R8 beeps; C7-R3 ↔ C8-R3 does not
☐	8 · Internal bridges	C11 R7-R8 and C7 R6-R7 beep; C7-R5 empty
☐	9 · External wires	purple/black/green identified; 1 kΩ in series with orange, with heat-shrink
☐	10 · Wire soldering	5 wires soldered, bridges C5, C7, C10 and bare wire C3 done
☐	11 · Multimeter	checks A–H beep, 1–8 do not, diode test ok, wires checked end to end
☐	12 · Pico	glue on the wires; Pico with USB towards C1
☐	Test 1	4.7 V on VSYS without USB; the PSU starts with GP8 (fan or 12 V on PCIe) and goes back to standby
☐	Test 2	pin 9 identified; the sensor reads on/off
☐	Test 3	full cycle with BOOTSEL
☐	Case	board insulated, antenna clear, old jumper removed, suspend disabled

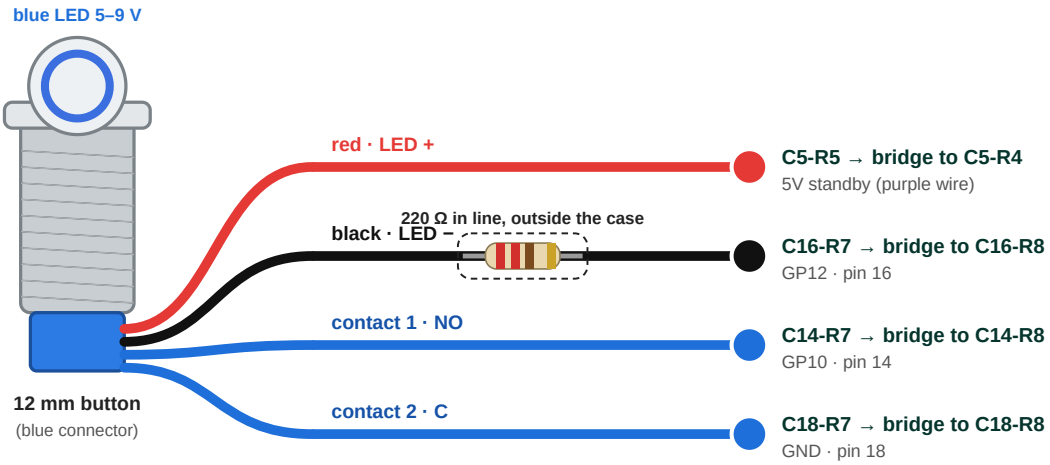
Power button with LED

Supplement to the Pico 2 W for BC250 manual · Gebildet 12 mm momentary button, blue LED 5–9 V, 4 wires

What you get	A button that behaves like a PC's: with the machine off one press turns it on, with the machine on a 4 s press forces it off. In addition, a double click with the machine on puts the dongle in pairing mode. The LED ring will be programmed later.
What you add	4 button wires, 4 solder bridges, one 220 Ω resistor in line on the black wire
Difficulty	Easy, if you have already built the board: 8–10 solder joints
Time	45 minutes + 15 minutes of testing
Version	1.2 · 28 September 2026 · resistor on the wire, behaviour aligned with the firmware

Where each wire goes

Button wire	What it is	Hole	Bridge to	Pico pin
red	LED +	**C5-R5**	C5-R4	5V standby (purple wire), not a Pico pin
black + 220 Ω in line	LED −	**C16-R7**	C16-R8	GP12 · pin 16
contact 1	NO	**C14-R7**	C14-R8	GP10 · pin 14
contact 2	C (common)	**C18-R7**	C18-R8	GND · pin 18



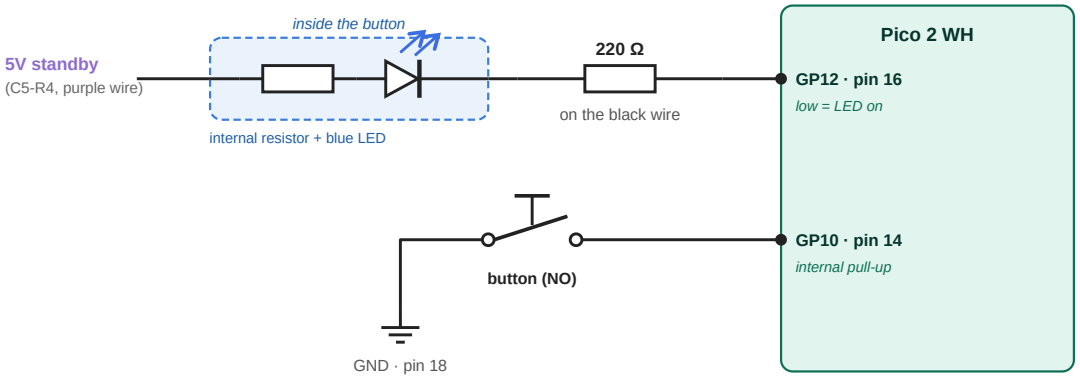
The two contact wires are identical: you can swap them. Red and black you cannot.

The wiring at a glance. The button wires go in from the copper side, like the other external wires.

Why these holes

The three wires on row R7 sit on **alternate** columns (14, 16, 18): there is always an empty column between them, so it is hard to join two bridges by mistake. The LED uses **GP12** and the ground is **pin 18**: any free GPIO and GND would work, these leave the most room for soldering.

1 How it works



Pressed: GP10 goes to ground and reads 0. Released: the pull-up holds it at 1.

Button schematic. The LED and the contact are two separate circuits inside the same button.

The LED

The LED takes 5 V from **5V standby** (purple wire, before the diode) and returns to ground **through GP12**. When the Pico drives GP12 **low** current flows and the LED lights; when it drives it high the LED goes off. This is called **active low**.

With PWM (GP12 switching on and off a thousand times a second) the Pico decides how long it stays low, and therefore the brightness: that is how it does breathing, blinking and steady light. In code the value is inverted: `duty 0` = full brightness, `duty 65535` = off.

The contact

The contact is a simple **normally open** (NO) switch between GP10 and ground. The Pico's internal pull-up holds GP10 at 1; when you press, GP10 goes to ground and reads 0. No other components are needed.

The 220 Ω resistor

The button already has an internal resistor for the LED, sized for 5–9 V, but its value is not stated. The 220 Ω in series on the black wire keeps the current into GP12 below 10 mA anyway, within what a Pico pin can handle. With the 5 V of 5V standby the LED is still clearly visible.

It goes on the wire, **outside the case**, 3–5 cm after the cable exit: at the exit the case must grip the bare wire, not the heat-shrink of the resistor.

2 Step-by-step assembly

What you need

Button	1 Gebildet 12 mm button with its 4-wire harness (the blue connector stays on the button)
Resistance	one 220 Ω, 1/4 W (red-red-brown-gold); anything from 150 to 330 Ω is fine
Heat-shrink	2–3 cm of thin heat-shrink tubing for the resistor, more for insulating
Lead offcuts	4 small pieces for the solder bridges (as in the manual)
Tools	soldering iron, solder, flush cutters, wire strippers, multimeter, magnifier

Before soldering

Unplug the PSU from the wall and take the Pico out of its headers. It goes back in only for the tests.

Step 1 Identify the wires

You need: button with harness, multimeter · Time: 5 minutes

1. Connect the harness to the button (blue connector).
2. Of the four wires, **red** and **black** are the LED: red = +, black = −.
3. The other two are the contact. Multimeter in **continuity**, one probe per wire: it must **not** beep at rest and must beep **only while you press**. They are identical: call them contact 1 and contact 2.
4. If instead it beeps at rest and stops when you press, you have an NC button: stop here, this guide needs a normally-open one.
5. Put a piece of tape on each wire marked **LED+**, **LED−**, **K**, **K**: in the case they get mixed up with the other wires.

Contact wire colours

The colour of the two contact wires changes from batch to batch. In the drawings they are **blue**: all that matters is that they are the two that beep when you press.

Step 2 Cut and prepare the wires

You need: flush cutters, wire strippers, solder · Time: 10 minutes

1. Decide where the button will go on the case and measure the route to the board. Cut the four wires to that length **plus 5 cm**.
2. Strip **2 mm** and tin the tip of each one: the wire is stranded, once tinned it becomes stiff and fits the hole.
3. Twist the four wires loosely together (one turn every 2–3 cm): they stay tidy and do not snag.
4. You fit the resistor on the black wire at step 4, after trying the route in the case.

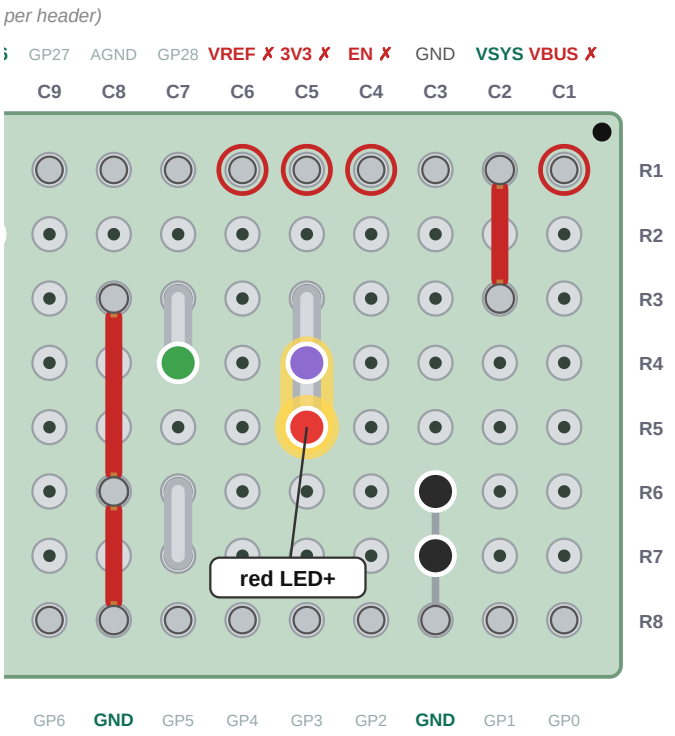
Does not fit the hole?

If the tinned tip is too thick, remove a little solder with desoldering braid. Do not enlarge the hole.

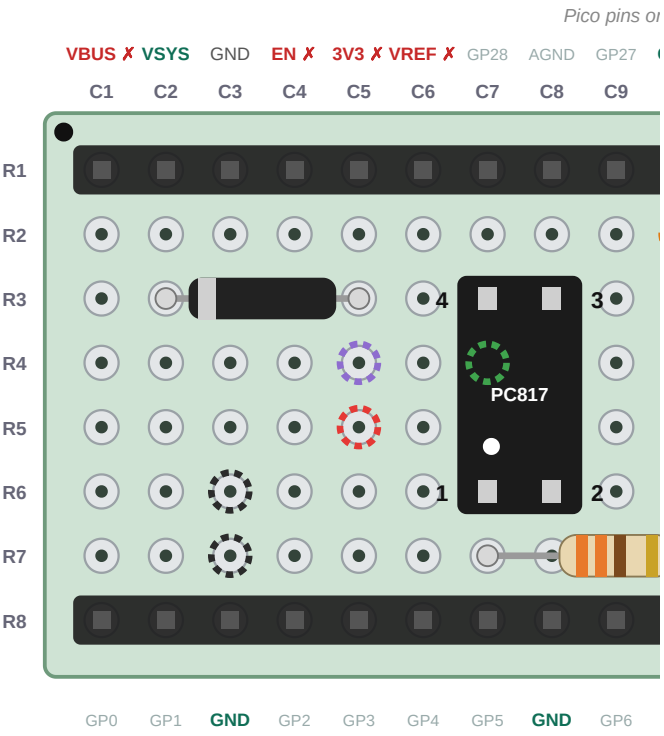
Step 3 Red (LED +) in C5-R5

You need: red wire, one lead offcut · Time: 5 minutes

Copper side (where you solder) · C1 on the right



Component side · C1 on the left



C5-R5 is the hole right below the purple wire (C5-R4), in the same column.

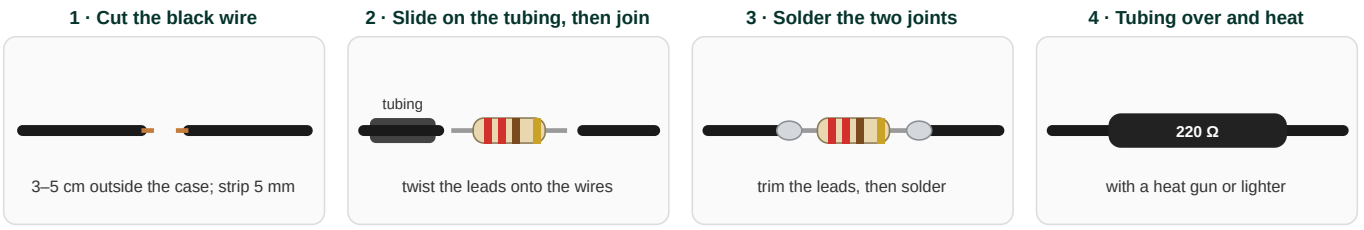
1. Push the tip of the **red** wire into **C5-R5** from the copper side, about 1 mm, and solder it.
2. Make the **bridge C5-R5 ↔ C5-R4** with a lead offcut: this gives the red wire the 5V standby of the purple wire.
3. On the component side the tip must not stick out: if it does, trim it flush.

Column 6 stays empty

Next to C5-R5 is **C6-R5**. If solder reaches it and then touches column 7, 5V standby ends up on PS_ON and the PSU turns on by itself. Check with the multimeter at step 6.

Step 4 220 Ω resistor on the black wire

You need: 220 Ω resistor, heat-shrink tubing · Time: 10 minutes



The 220 Ω in series on the black wire, protected by heat-shrink: same method as the 1 kΩ on the orange wire.

1. Identify it: **red-red-brown-gold** (5-band: red-red-black-black-brown). If in doubt measure it: 215–225 Ω.
2. From hole C16-R7 to the cable exit of the case (right-hand side, looking at the USB) the black wire runs about 2 cm. Mark the black wire with a marker **6 cm from the tinned tip**: this puts the resistor about 4 cm outside the case.
3. Cut the black wire at the mark, strip 5 mm on both ends. **Slide the heat-shrink on first**, then solder the resistor between the two pieces, twisting its leads onto the wires.
4. Trim the leads, slide the heat-shrink over and heat it.
5. Check in Ω: between the two ends of the black wire you must read about 220 Ω.

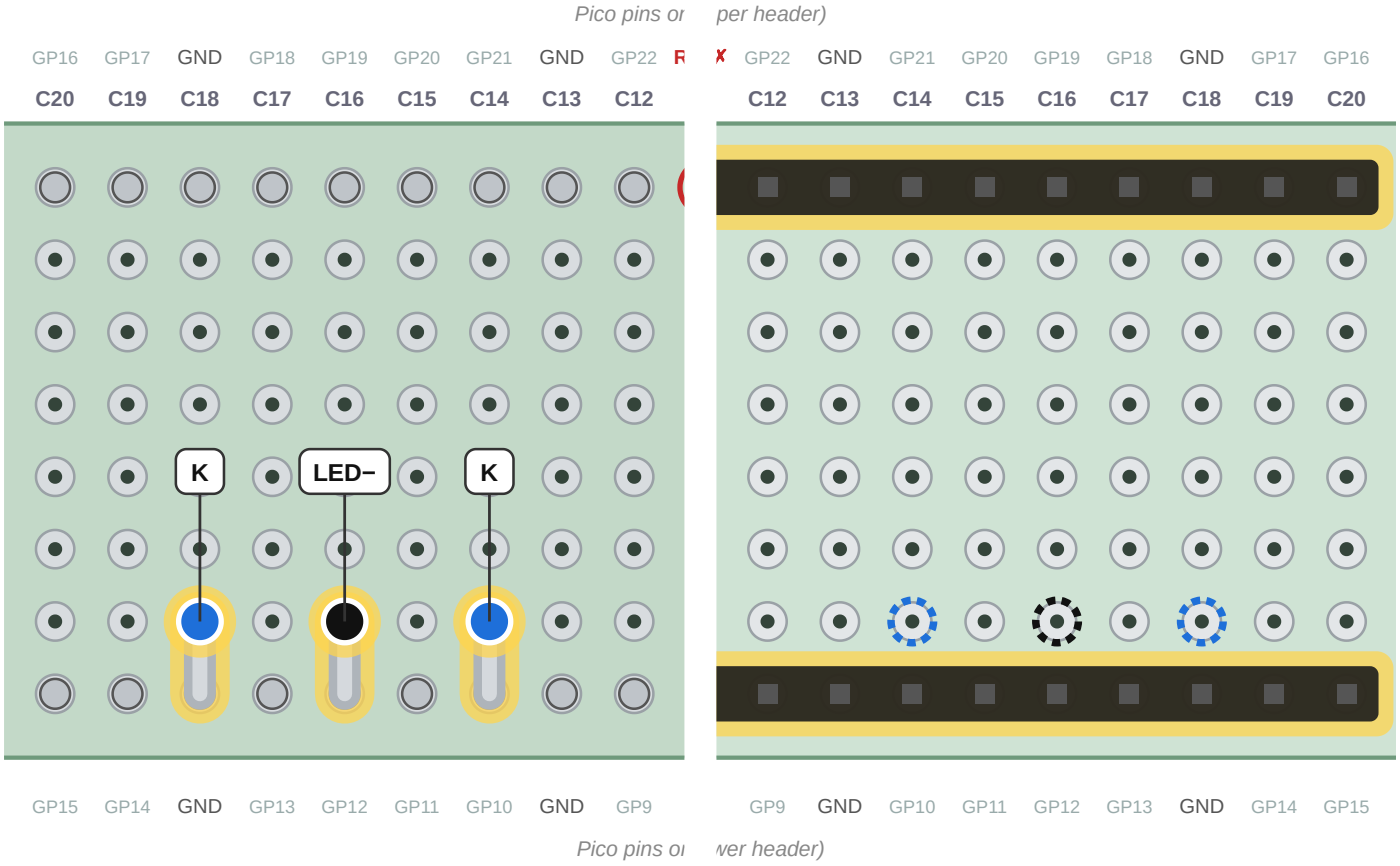
Not inside the cable exit

The resistor and its heat-shrink are thicker than the wire and do not fit the groove. If they end up inside, the case does not close or the wire is not gripped.

Step 5 Contact and LED − on row R7

You need: black wire and the two contact wires, 3 lead offcuts · Time: 15 minutes

Copper side (where you solder) · C20 on the left — **Component side · C20 on the right**



The three new joints are on row R7, just inside the lower header (R8). One empty column between each.

1. **Contact 1** in **C14-R7**, then **bridge C14-R7 ↔ C14-R8** (pin 14 = GP10).
2. **Black** (LED −, with the resistor already fitted further along the wire) in **C16-R7**, then **bridge C16-R7 ↔ C16-R8** (pin 16 = GP12).
3. **Contact 2** in **C18-R7**, then **bridge C18-R7 ↔ C18-R8** (pin 18 = GND).
4. Each tip goes in from the copper side about 1 mm and must not stick out on the component side: the Pico header is there.
5. **Vertical**, narrow bridges: little solder, just enough to join the two holes.

Mind the neighbours

Between the holes used, C15 and C17 stay free on row R7, but on row R8 there are the header pins GP11 (C15) and GP13 (C17). Solder that slips sideways joins two Pico pins: step 6 catches it.

Step 6 Check with the multimeter

You need: multimeter · Time: 10 minutes · Pico NOT inserted, PSU unplugged

1. Multimeter in **continuity**. The green pairs (letters) must beep, the red ones (numbers) must not.
2. If a red one beeps, look for the excess solder with the magnifier and remove it with braid.
3. Also repeat the checks of step 11 of the manual: if anything flowed onto the old connections, you find out now.

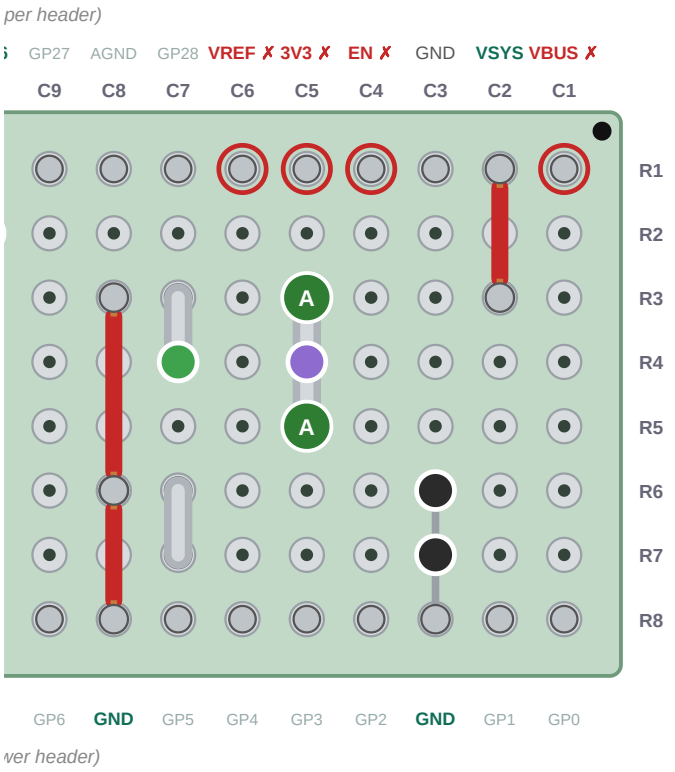
Must beep

	Between	and
A	C5-R5 (LED+)	C5-R3 (D1 anode)
B	C14-R7	C14-R8 (GP10)
C	C16-R7	C16-R8 (GP12)
D	C18-R7	C18-R8 (GND)
E	C14-R8, **button pressed**	C18-R8

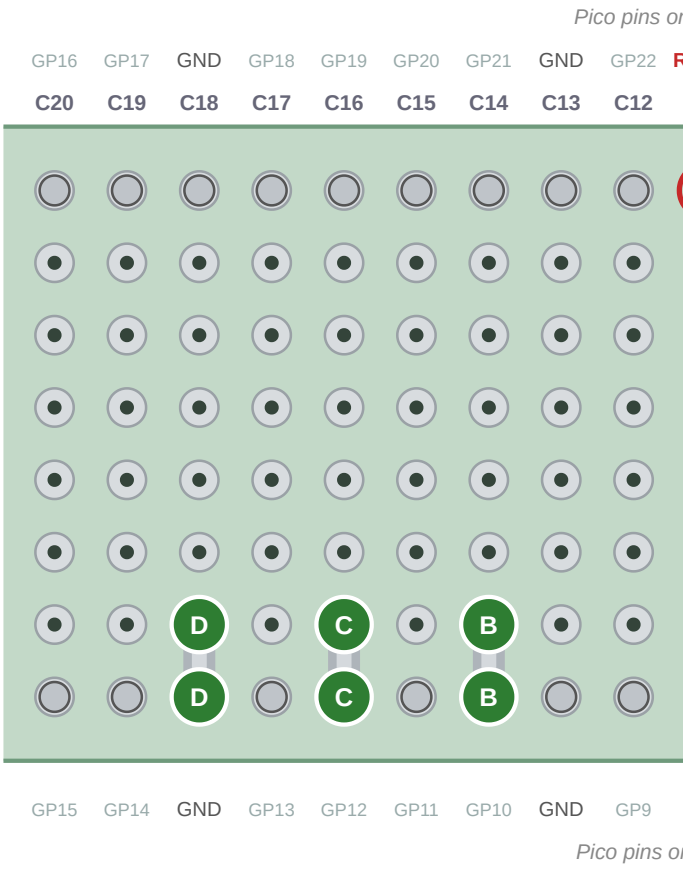
Must not beep

	Between	and
1	C5-R5 (LED+)	C6-R5 and C4-R5
2	C5-R4 (5VSB)	C7-R4 (PS_ON)
3	C5-R5 (LED+)	C3-R8 (GND)
4	C14-R8	C13-R8 and C15-R8
5	C16-R8	C15-R8 and C17-R8
6	C18-R8	C17-R8 and C19-R8
7	C14-R8, button **released**	C18-R8

Copper side · columns 1–9



Copper side · columns 12–20



The points that must beep with each other. For E hold the probes on C14-R8 and C18-R8 and press the button.

The 220 Ω, end to end

In Ω, between the tip of the black wire at the blue connector end (unplugged from the button) and C16-R8: about 220 Ω.

Step 7 Refit the Pico and close the case

You need: hot glue, the case fasteners · Time: 10 minutes

1. A drop of hot glue where the four wires leave the board, on the copper side: this way, if the button gets pulled, the solder joints do not.
2. Plug the Pico back in with the USB towards C1, as always.
3. Rest the board on its supports in the bottom half of the case and lay each wire in its own cable exit groove, one wire per groove.
4. The resistor on the black wire stays outside, 3–5 cm after the exit.
5. Close the case: the two halves grip the wires.

> **Hole in the case**
>
> The button needs a **12 mm** hole (printed: 12.2–12.3 mm) and a wall of at most 5–6 mm for the nut. If the hole in the case is larger (for example 16 mm), you need a reducing ring.

3 MicroPython tests

With MicroPython and Thonny, as in the manual. Connect the Pico to the PC over USB and **also plug in the PSU**: the LED takes its 5 V from 5V standby, not from USB. GP8 is not touched, so the PSU stays in standby.

Test 1 · the contact

```
from machine import Pin
import time

button = Pin(10, Pin.IN, Pin.PULL_UP)   # pressed = 0

while True:
    print("pressed" if button.value() == 0 else "released")
    time.sleep(0.2)
```

Correct result: “released” at rest, “pressed” for as long as you hold the button down.

Test 2 · the breathing LED

```
from machine import Pin, PWM
import math, time

led = PWM(Pin(12), freq=1000)

def light(x):                          # x from 0 (off) to 1 (full)
    led.duty_u16(int(65535 * (1 - x)))  # active low: inverted

t = 0
while True:
    light((1 - math.cos(t)) / 2)
    t += 0.05
    time.sleep(0.02)
```

Correct result: the ring fades in and out slowly, about every 2.5 s.

Test 3 · everything together

```
from machine import Pin, PWM
import math, time

button = Pin(10, Pin.IN, Pin.PULL_UP)
led = PWM(Pin(12), freq=1000)

def light(x):
    led.duty_u16(int(65535 * (1 - x)))

t = 0
while True:
    if button.value() == 0:           # pressed: steady at full
        light(1)
    else:                             # released: breathing
        light((1 - math.cos(t)) / 2)
        t += 0.05
    time.sleep(0.02)
```

Correct result: the LED breathes; while you hold the button it stays steadily on. If the LED never lights, see chapter 5.

Test 4 · press recognition

Same thresholds as the final firmware. It does not touch GP8, so the PSU does not start: it only prints what it recognised.

```
from machine import Pin
import time

button = Pin(10, Pin.IN, Pin.PULL_UP)
DEBOUNCE = 30       # ms
SHORT = 1000        # ms: beyond this it is no longer a click
DOUBLE = 400        # ms between the first release and the second press
LONG = 4000         # ms: forced power-off

prev = False; t_change = 0; t_down = 0; t_up = 0
clicks = 0; long_done = False

while True:
    now = time.ticks_ms()
    p = button.value() == 0
    if p != prev and time.ticks_diff(now, t_change) > DEBOUNCE:
        prev = p; t_change = now
        if p:                                   # pressed
            t_down = now; long_done = False
        elif not long_done:                     # released
            if time.ticks_diff(now, t_down) < SHORT:
                clicks += 1; t_up = now
                if clicks == 2:
                    print("double click -> pairing"); clicks = 0
            else:
                print("medium press -> no action"); clicks = 0
    if p and not long_done and time.ticks_diff(now, t_down) >= LONG:
        print("held 4 s -> forced power-off"); long_done = True; clicks = 0
    if clicks == 1 and not p and time.ticks_diff(now, t_up) > DOUBLE:
        print("single click -> reserved"); clicks = 0
    time.sleep_ms(5)
```

Correct result: one click prints “single click”, two quick clicks “double click”, holding for 4 s shows “forced power-off” while the button is still down.

4 What the firmware will do

The final firmware is written by Claude Code from the project prompt (a fork of DS5Dongle with the power bridge). This is only the summary, so you know what to expect when you try the button.

Pinout

Pin	Function
GP8 · pin 11	output to the PC817 → PS_ON (high = PSU on)
GP26 · pin 31	input from TPMS1 pin 9 (high = BC250 on)
GP10 · pin 14	button, internal pull-up, pressed = 0
GP12 · pin 16	button LED, active low; off for now

The button

Machine state	Press	What happens
Off	any	it powers on
On	held for 4 s	forced power-off, then a 15 s pause
On	double click (within 400 ms)	the dongle enters pairing for 30 s
On	single click	no action for now (reserved)
On	between 1 and 4 s	no action
Powering on, or in the 15 s pause	any	ignored

The controllers

Event	What happens
Machine off, a paired DualSense connects (PS button)	the machine powers on
Machine off, a new controller is paired (BOOTSEL once)	when pairing completes the machine powers on
The machine turns off, for any reason	all controllers are disconnected (pairings are kept)
In the 15 s after shutdown	connections do not power the machine back on

Forced power-off is like pulling the plug

Use it only if Bazzite is frozen: it cuts power to the BC250 without shutting the system down. To shut down normally use the Steam menu.

5 Troubleshooting

Symptom	Likely cause	What to do
The LED never lights	PSU unplugged; bridge C5 R4–R5 missing; red and black swapped	Test 2 with the PSU plugged in; check A of step 6; resistor end to end (step 6)
The LED is always on at full brightness	Black wire soldered to a GND instead of GP12; C16 bridge touching C15 or C17	Checks C and 5 of step 6
The LED is dim even at full	Resistor value too high (for example 2.2 kΩ: red-red-red)	Check the colours or measure it: it must be 150–330 Ω
The case does not close on the button wires	Resistor or heat-shrink ended up inside the cable exit	Move the resistor 3–5 cm outside the case
Test 1 always says “pressed”	C14 bridge touching C13 (GND) or C15; NC contact instead of NO	Checks 4 and 7 of step 6; step 1
Test 1 always says “released”	C14 or C18 bridge missing; contact unplugged at the blue connector	Checks B, D and E of step 6
The PSU turns on by itself	Solder between C5 and C7 across column 6	Check 2 of step 6
The Pico reboots when you press	The contact pulls the wrong pin to ground (for example RUN)	Recheck the columns: the contact must reach C14 and C18, row R8

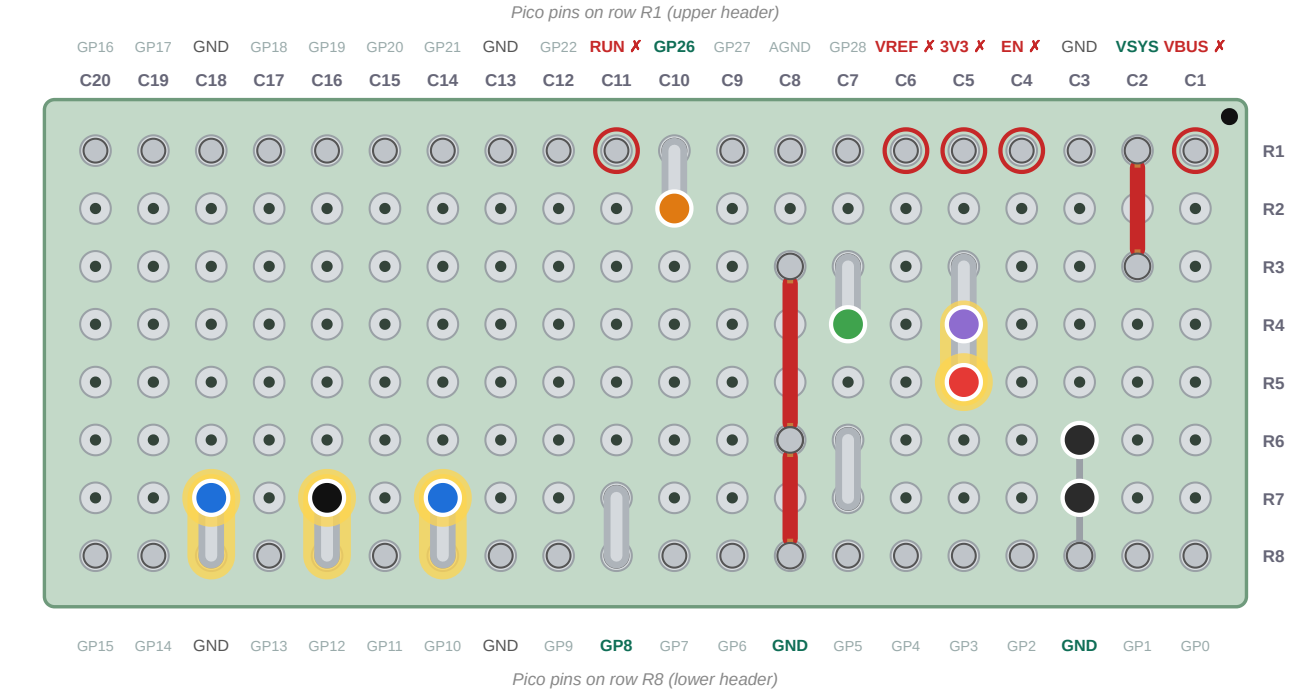
Summary table

Connection	Path
LED +	red wire → C5-R5 → bridge → C5-R4 (purple wire, 5V standby)
LED −	black wire → 220 Ω in line (outside the case) → C16-R7 → bridge → C16-R8 (GP12, pin 16)
Contact 1	C14-R7 → bridge → C14-R8 (GP10, pin 14)
Contact 2	C18-R7 → bridge → C18-R8 (GND, pin 18)
Holes to leave empty	C4-R5, C6-R5, C5-R6 · C13, C15, C17 and C19 on row R7

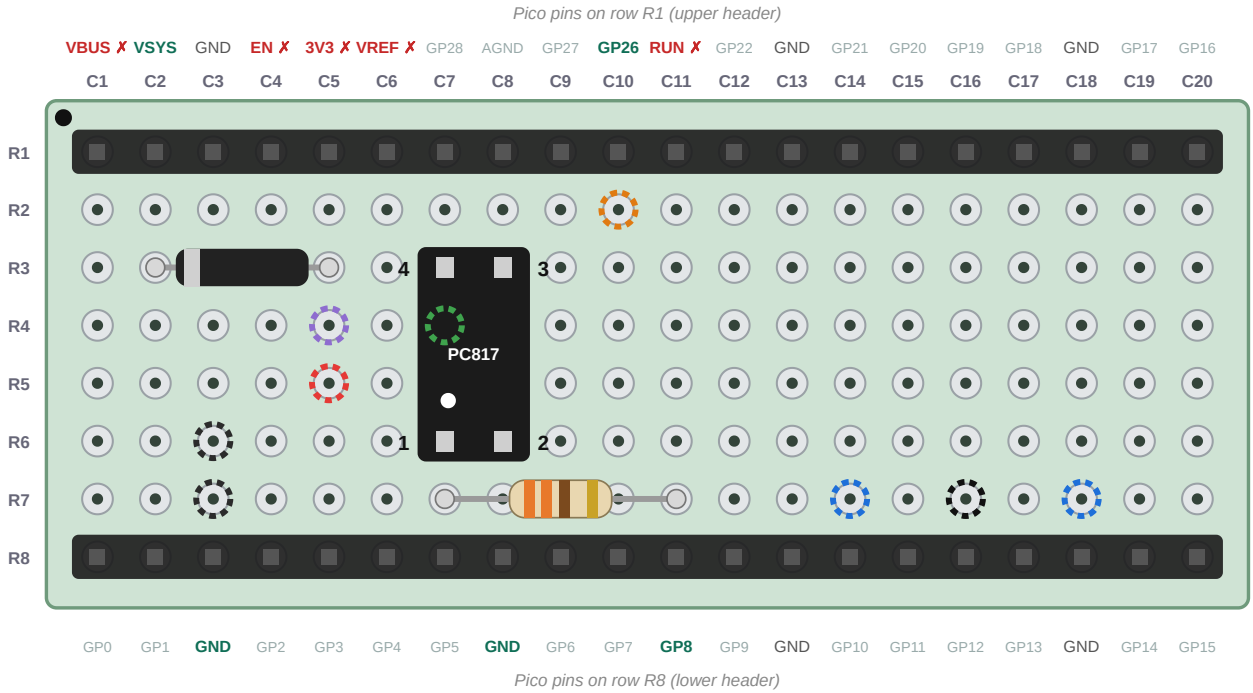
Checklist

✓	Step	Done when…
☐	1 · Wires	LED+, LED−, K, K labelled; the contact beeps only when pressed
☐	3 · Red	C5-R5 soldered, bridge to C5-R4, C6-R5 empty
☐	4 · 220 Ω	on the black wire, 3–5 cm outside the case, heat-shrink shrunk, 220 Ω measured
☐	5 · Row R7	C14, C16, C18 soldered with their bridges to R8
☐	6 · Multimeter	A–E beep, 1–7 do not, 220 Ω end to end
☐	7 · Case	wires in their cable exits, resistor outside, case closed
☐	Tests	tests 1, 2, 3 and 4 passed

A Complete board with the button



Copper side, all connections. The button's are in yellow.



Component side. The dashed circles are the wires coming in from below.